CHAPTER I

The Problem and Its Background

Introduction

The development of internet and information technology made a system to the practical, effective, and efficient. One of the impacts is ease to make scheduling information system in university Khoirul Anam (2020). This shows how technology can simplify complex administrative tasks and improve overall productivity.

This study focuses on designing a web-based scheduling system for Rooms, Teachers, and Section. It aims to streamline the scheduling process by automating room assignments, teacher allocations, and section timetables, ensuring accuracy, efficiency, and accessibility for both administrators and users. According to Jesus Abraham Garcia Yanez et al. (2024) Efficient scheduling in academic institutions is crucial to ensure the optimal utilization of resources, meet the diverse needs of students, and facilitate the smooth functioning of educational programs. However, manual scheduling processes are often difficult and error-prone.

Through this system, administrators can easily assign rooms, arrange class schedules, and manage teacher workloads without confusion or conflicts. Teachers can quickly check their schedules, while students can view their class timetables online, making it more convenient than printed schedules.

Background of the Study

In the case of ACLC College of Taytay this process is currently handled manually, often through spreadsheets and manual checking. While this approach works to some extent, it is time-consuming, prone to errors, and vulnerable to conflicts such as double bookings, uneven teacher workloads, and underutilized classrooms. As a result, students sometimes experience inconvenience, such as having to move from one room to another due to incorrect teacher assignments and improper room allocations.

This challenge is not only experienced at ACLC College of Taytay but also encountered in other academic institutions. According to Jose C. Agoylo Jr. et. al (2025). Manual scheduling methods are prone to conflicts, delays, and inconsistencies due to limited adaptability to institutional constraints. This innovative solution leverages advanced technologies to streamline scheduling, reduce conflicts.

To address these challenges, A web-based scheduling and assignment system would be highly beneficial for ACLC College of Taytay. Such a system can help streamline the scheduling process, reduce errors, and ensure that academic activities run smoothly for both students and teachers.

General Objective The objective is to create a Web-Based Scheduling and Assignment System for Rooms, Teachers, and Sections at ACLC College of Taytay that will make the scheduling process easier, faster, avoid conflicts and more reliable for the school.

Specific Objective To develop a Web-Based Scheduling and Assignment System for Rooms, Teachers, and Sections at ACLC College of Taytay, aims to solve the challenges of the current manual scheduling process.

Specifically, this study aims to:

1. To Design and Create/Develop

1.1 A secure web-based scheduling system that features user authentication and role-based access for administrators, teachers, and students.

1.2 Provide the administrator with full management control to add, edit, delete, and view records of teachers, rooms, subjects, and sections.

1.3 Generate scheduling module that allows the administrator to assign teachers, subjects, and rooms to class sections while detecting time and room conflicts.

1.4 Enable teachers and students to view their respective schedules through personalized “My Timetable” that display subject, time, and room assignments.

1.5 Implement a notification feature to inform users of updates, changes, or new schedules in real time.

2.Test the functionality and performance of the system to verify that all modules, features, and processes operate accurately and efficiently.

3.To evaluate the system using ISO25010. In terms of Functionality, Performance Efficiency, Usability, Compatibility, Reliability, Security, Maintainability and Portability.

Scope

User Authentication and Role Management

- This feature ensures that only authorized users can access the system. Each user must log in using a valid username and password.
- The system includes three roles: Administrator, Teacher, and Student, each having different access levels and permissions.

User Management

- The User Management module allows the Administrator to handle all system accounts.
- The admin can create new user accounts for teachers and students, assigning them their appropriate roles and login credentials.
- The admin inputs the following information for teacher and student accounts:
    - For teachers, the administrator inputs information such as the teacher's Full name, Email, Username, Password, Employee ID, Department, Account status (active or inactive), subjects handled (used for assigning schedules) Expertise.
    - For students, the administrator inputs information such as the student's Full name, Email, Username, Password, Student ID,

Program, account status (active or inactive), year level or grade, and section.

- The admin can edit user profiles, such as:
  - Full Name, Username, Password, Employee ID, Department, Account Status, Subjects Handled, Student ID, Course / Strand, Year Level / Grade, and Section.
- The admin can activate, deactivate, or delete accounts depending on user status (e.g., graduated student, resigned teacher).

Rooms, Teachers, Sections, and Subjects Management

Rooms Management

- Admin can add, view, update, and delete rooms.
- Admin can edit room information, including:
  - Room name, Capacity, Building, Floor, Room type (lecture, laboratory), and description.

Section Management

- Admin can add, view, update, and delete sections.
- Admin can edit section information, including:
  - Section name, Program Type, Year Level, Capacity, Term, Curriculum, Status, Adviser.

Subject Management

- Admin can add, view, update, and delete subjects.
- Admin can edit subject information, including:
  - Subject name, Subject code, Lecture Unit, Labaratory Unit, Required room type, Status, Description.

Teacher Management

- Admin can add, update, and delete teacher records.
- Admin can edit teacher information, including:
  - Employee ID, Department, Subjects handled, sections assigned, and Expertise.

Create and Assignment Schedule

- The Schedule Module is the core feature that allows the administrator to assign teachers, subjects, rooms, and sections.
- Admin can create, update, and delete schedules according to available time slots and days.
- During schedule creation, the system automatically checks for conflicts such as:

    - Teacher Conflict – when a teacher is assigned to two different classes at the same time.
    - Room Conflict – when a room is scheduled for multiple classes simultaneously.
    - Section Conflict – when a section is assigned to overlapping subjects.

- Schedule information includes:
    - teacher, subject, section, room, day, time start and end, and school year or semester.
- The system ensures that no duplicate or overlapping schedules are saved.
- The administrator can view schedules by:
    - Teacher, Section, Room
- Notifications are automatically sent to teachers and students if there are schedule updates or changes.

Curriculum Management

- Admin can manage the curriculum structure used for scheduling and subject assignments.
- Includes:
- School Year Start, Program Type, Program, Type.
- Ensures that only valid subjects within the selected curriculum are used in schedule creation.
- Supports updates or revisions to curriculum details when new academic programs are added.

Term Management

- Admin can define and control the academic term and school year used in scheduling.
- Includes:
- Term code, Academic Year, Semester, Status.
- Ensures that schedules are created only for the active term set by the administrator.

Notification

- The administrator can post announcements or schedule changes, which appear as notifications on the user dashboards.
- Teachers and students are informed of any schedule adjustments through the system's notification feature.

Delimitation

User Access Limitation

- The system is designed only for authorized users such as administrators, teachers, and students within the institution.
- It does not allow guest access or registration without admin approval.
- Password recovery through email or SMS is not yet implemented and
- must be done manually by the administrator.

Internet and Server Dependency

- Since the system is web-based, it requires a stable internet connection and a server (e.g., Laragon or online hosting) to function properly.
- It will not operate offline unless configured locally.

No Automated SMS or Email Notification

- Notifications are limited to the system dashboard interface only.
- The system does not send text messages or emails for schedule updates or announcements.

Limited Scheduling Scope

- The system focuses solely on academic scheduling (rooms, teachers, subjects, and sections).
- It does not include extracurricular event scheduling, meeting room reservations, or facility maintenance schedules.

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Curriculum and Data Entry

- All curriculum, teacher, student, subject, and room information must be manually encoded by the administrator.
- The system does not automatically import data from external databases or enrollment systems.

Conflict Detection Scope

- The system checks for basic time conflicts (teacher, section, and room overlaps).

Platform Compatibility

- The system is optimized for desktop, mobile and laptop browsers

Single Institution Use

- The system is intended for use within one school or campus only.
- It does not support multiple institutions or cross-campus scheduling integration.

Significance of the Study

This study is important because it shows how the proposed Web-Based Scheduling and Assignment System can make school tasks easier and more organized. It focuses on helping the people who are directly involved in scheduling: the administrators, teachers, students, and even future researchers.

Administrators

The system makes it easier to edit and manage all schedules. They can assign teachers, rooms, and sections more accurately, and they can also make quick changes whenever needed. This reduces errors and saves a lot of time.

Teachers

The system provides a clear view of their schedules. They no longer have to worry about overlaps or last-minute changes because updates are shown right away. This helps them focus more on teaching instead of dealing with scheduling issues.

Students

The system allows them to view their schedules anytime. It lessens confusion and makes sure they are always updated. With this, students can prepare better for their classes and manage their time more effectively.

Researchers

This study is significant to the researchers as it enhanced their technical and analytical skills in developing a functional web-based scheduling and assignment system. It also contributed to their growth not only as developers but as researchers who understand the importance of connecting concepts from existing studies to practical, real-life applications. Through this study, they were able to apply their academic knowledge, think critically, and develop a meaningful system that can provide practical benefits to others.

Future Researchers

This study can serve as a reference and guide. It may help them develop similar systems, improve existing features, or explore new innovations in school management and scheduling.

Definition of Terms

Web-Based Scheduling System – A computer-based application accessible through a web browser that automates and organizes the scheduling of rooms, teachers, and student sections in an academic institution.

PHP (Hypertext Preprocessor) – An open-source scripting language commonly used for web development. In this study, PHP serves as the server-side language responsible for processing system logic and managing database transactions.

HTML (Hypertext Markup Language) – The standard markup language for creating and structuring the content of web pages. It is used in this project to design the framework of the system's interface.

Tailwind CSS – It allows developers to apply predefined classes directly in the HTML to style elements without writing custom CSS. In this study, Tailwind CSS was used to create a clean, responsive, and consistent design for the system's interface, making it easier for users to navigate and improving the overall user experience.

JavaScript – A client-side programming language that enables interactivity and dynamic functions on web pages. In this system, JavaScript is used to provide features such as real-time updates and responsive user interactions.

Laravel – A PHP-based web framework that follows the Model-View-Controller (MVC) architecture. It is applied in this project to streamline system development, enhance security, and ensure scalability and maintainability.

Laragon – A lightweight and portable local development environment used for web application development. It provides a complete package that includes Apache, MySQL, PHP, and other tools necessary to run and test web-based systems on a local computer. In this study, Laragon is used as the local server environment for developing, running, and testing the Web-Based Scheduling System before deployment.

Database – A structured digital repository where relevant data, such as room information, teacher schedules, and section assignments, are stored, retrieved, and managed.

CHAPTER II

Review of Related Literature and Studies

This chapter reviews foreign and local literature and studies relevant to web-based scheduling for rooms, teachers, and sections to position the present study within existing knowledge and identify gaps the system will address.

Foreign Literature

Development of a Web-Based Timetabling Software for a Mexican University

According to Garcia-Yañez et al. (2024), the process of creating class schedules in universities is often time-consuming and prone to errors, especially when done manually. Their study focuses on developing a web-based timetabling software that aims to simplify and automate the scheduling process in a Mexican university. The researchers designed a web application that allows administrators and teachers to input data such as class schedules, room assignments, and teacher availability. The system automatically checks for possible conflicts, such as overlapping classes or unavailable rooms, and provides a visual interface where users can easily manage and adjust timetables. The software follows a structured web-based architecture composed of a front-end, back-end, and database. The front-end focuses on providing an easy-to-use interface for teachers and administrators, while the back-end manages data processing and schedule validation. The system also uses Firebase, a cloud-based database, to store and manage real-time scheduling data. Through testing, the study showed that the developed system was able to reduce human error, speed up scheduling tasks, and make timetable management more efficient and organized.

Synthesis

Their Study helps our study by showing the effectiveness of using web-based systems to automate and simplify the scheduling process. Their system serves as a valuable reference for designing the interface, structure, and conflict-handling functions of the proposed web-based scheduling and assignment system for rooms, teachers, and sections.

Modelling and solving the university course timetabling problem with hybrid teaching considerations

Davison Kheiri, et al. (2024) conducted a study that focuses on improving the scheduling process in universities, especially with the growing use of hybrid learning. The researchers explained that timetabling has always been a difficult task since it involves arranging classes, teachers, and rooms without conflicts in time and resources. With the rise of hybrid teaching—where some classes are held online, in-person, or both—the process has become even more challenging. To address this, they developed a multi-objective programming model that considers important factors such as classroom availability, instructor schedules, and student preferences. The model also includes hybrid teaching features to make the schedule more flexible and efficient. Using a step-by-step method, it helps prioritize different goals, such as reducing schedule conflicts and improving the use of university facilities. The results of their experiments showed that the model can create balanced and demand-based timetables that work well for hybrid learning setups.

Synthesis

This study provides helpful insights for the researchers' system since it presents advanced methods of optimizing schedules through automation and logical modeling. It also focuses on hybrid teaching, its approach to organizing classes, teachers, and rooms efficiently can serve as a reference for designing the scheduling flow in our proposed web-based system.

Class Scheduling Web Application Using Django Framework

The study by Anubhav Rawal (2021) at Western Michigan University aimed to develop a modern web-based scheduling system using the Django framework. The project was designed to help the Computer Science Department manage class schedules, faculty assignments, and classroom allocations more efficiently. Previously, scheduling was done using spreadsheets and an outdated Ruby on Rails application, which was functional but lacked flexibility and ease of use. The new Django-based system introduced features such as uploading Excel files for schedule data, dynamic editing of multiple entries, conflict checking for overlapping classes, and user roles with different access levels (admin, teacher, student). It also included a recycle bin for deleted data recovery and was built with scalability in mind for future department expansion. Although some functions like user authentication and full conflict detection were not completed due to time limits, the system successfully provided a strong foundation for future improvements. Overall, the project demonstrated how web-based tools can enhance scheduling efficiency, reduce manual errors, and improve the overall workflow in academic institutions.

Synthesis

Their system used Django, the key ideas such as importing class data, assigning teachers and rooms efficiently, and minimizing scheduling errors are directly relevant to our study. It highlights the importance of automating the scheduling process to replace manual methods like spreadsheets, helping reduce conflicts and save time. This supports our system's goal of creating a more organized, efficient, and user-friendly scheduling process for ACLC College of Taytay.

Automated Large-scale Class Scheduling in MiniZinc

According to Rahman et. al (2020), their study focuses on automating large-scale class scheduling in universities using the Mini Zinc modeling language together with ready-made solvers. They describe the scheduling process as a complex task that involves assigning courses, teachers, rooms, and time slots without causing overlaps or conflicts, while also ensuring that resources are used efficiently. Their proposed model was initially designed for a fixed credit system, where students follow a set program of courses, but they emphasize that it can also be adjusted to fit more flexible academic structures. Their study incorporated two types of scheduling rules — hard constraints, which must always be followed (such as preventing overlapping classes or double-booked rooms), and soft constraints, which are desirable but can be adjusted if necessary (like preferred class times or instructor choices). Using the Mini Zinc modeling language and built-in solvers, the researchers were able to automatically generate schedules that effectively balanced these conditions. Remarkably, their system could create organized and conflict-free timetables for mid-sized universities in less than a minute, showing its strong potential for practical academic use.

Synthesis

Their study focused on creating an automated scheduling model that efficiently generates complete semester timetables using Mini Zinc solvers. Our study builds on this foundation by translating the idea of automated scheduling into a fully interactive, web-based platform that can be utilized by both technical and non-technical users.

Explicit Artificial Intelligence Timetable Generator for Colleges and Universities

According to Lawrence A. Farinola et al. (2025), creating college and university timetables is often a challenging and error-prone task when done manually. The researchers developed an AI-based timetable generator that uses a Genetic Algorithm to automatically create schedules. The system considers important factors like room availability, lecturer assignments, and avoiding course clashes, ensuring that the resulting timetable is practical and conflict-free. By testing the system with real institutional data, the researchers found that it could produce schedules much faster and more accurately than traditional methods. Beyond just saving time, AI can make administrative tasks in schools more efficient. The system doesn't simply assign courses randomly—it optimizes the schedule so that classrooms, lecturers, and students are all considered, making the process smoother and fairer for everyone involved. This demonstrates the potential of AI to handle complex, repetitive tasks, freeing staff to focus on more meaningful work and helping institutions manage their resources better.

Synthesis

This study can guide our research by showing how AI and Genetic Algorithms can efficiently create conflict-free and optimized timetables, providing a strong foundation for developing our own scheduling system. In turn, our study adds value to theirs by implementing a practical, web-based interface for rooms, teachers, and sections, making the AI-driven scheduling more interactive, accessible, and usable in real-world settings. Together, these approaches highlight how combining smart algorithms with user-friendly platforms can significantly improve the efficiency and effectiveness of academic scheduling.

Local Literature

Designing Of Web-Based Attendance and Class Scheduling System using RFID and Raspberry PI

According to Reyes et al. (2023), their study developed a web-based attendance and class scheduling system for Columban College in the Philippines. The project aimed to make attendance checking and class scheduling faster and more accurate, replacing the old manual process that was often slow and prone to mistakes. Their system used RFID technology and a Raspberry Pi device to automatically record attendance whenever a student tapped their RFID card. All attendance data was saved in a centralized database, which teachers and administrators could easily access in real time. The system also featured automatic report generation, schedule displays, and SMS notifications sent to parents and school staff to improve communication and transparency. The researchers followed the waterfall model in creating their system, which included steps such as planning, design, coding, testing, and maintenance. The results of their study showed that the system helped improve the accuracy and efficiency of attendance monitoring while reducing the workload of teachers. It also made data sharing and updates easier, allowing students, teachers, and administrators to stay informed through automated and real-time access to information.

Synthesis

This study supports our study by demonstrating how web-based systems can automate school processes efficiently. Their system inspired the integration of real-time data management and user-friendly scheduling features in the proposed study.

Automated Class Scheduling System for Aemilianum College Inc.

According to Sarmiento et al. (2025), their study focused on developing an automated class scheduling system for Aemilianum College, Inc. The researchers aimed to improve the traditional manual scheduling process, which was often time-consuming, prone to errors, and difficult to manage as the number of courses, teachers, and classrooms increased. Their system was designed to help administrators easily manage faculty loads, class schedules, and room assignments through an automated platform. The developed system included an Admin Module that allowed users to manage schedules, assign teachers, configure settings, and generate reports. It also provided features for viewing faculty schedules, classroom utilization, and subject assignments. The researchers evaluated their system using the ISO 25010 standard, which measures software quality in terms of functionality, performance, usability, security, and reliability. Based on the evaluation results, both IT experts and end users rated the system as functional, reliable, and efficient in improving the scheduling process of the college.

Synthesis

Their system demonstrated how digital tools can help administrators efficiently manage teacher loads, class schedules, and room assignments within one platform. Our study builds on these gaps by developing a web-based system that can handle more complex scheduling conditions, automatically detect conflicts, and allow users to access and update information anytime and anywhere.

Scheduling System for Teacher of Lagro High School

According to Tyrone Catubao et al. (2020) Computer technology is used in many different contexts for daily life in the current generation. Whether they realize it or not, everyone has to have a computer. To a degree, everyone uses technology in their daily lives. Examples include using gadgets or mobile phones to stay updated or setting an alarm clock. Technology is utilized in a variety of settings, including business, industry, education, healthcare, and research. Lagro Senior High School was offered a computerized scheduling system to help them quickly construct professor schedules. The purpose of the study was to identify a school where the scheduling method was difficult. The selected school was given a system by the researchers to replace its manual processes. The principal of Lagro Senior High School was interviewed by the researchers, and the interview went well. The researchers created a system to assist the school in solving its issues by utilizing the waterfall model and the systems development life cycle method. The primary issue at Lagro Senior High School is the ineffective scheduling of teachers' schedules; as a result of their manual scheduling technique, they are running out of time to teach. The security flow issue also affects Lagro Senior High School. As a result, anybody can alter the timetable. In order to address this issue, the researchers successfully introduced the suggested system to Lagro Senior High School and had it put into place there in a fully operational state. Important functions like database maintenance and login/logout services were included in the proposed system. The purpose of the login form was to protect system users' security. Additionally, the system included a login history feature that allows users to view their past account information. The system also includes backup data, which allows users to locate and recover accidentally erased files. The results showed that Lagro High School's scheduling strategy worked well. A new scheduling system that Lagro High School can use to create and generate classrooms and individual faculty members was developed as a result of this study. It also enables users to automatically detect and eliminate conflicts while establishing schedules.

Synthesis

This study demonstrates how the integration of computer technology can significantly enhance the efficiency and reliability of academic scheduling. Their study mainly focused on teacher timetables and did not include room or section management. Our study builds on this by developing a more comprehensive web-based scheduling and assignment system that integrates teachers, rooms, and sections into one platform, providing real-time access and online updates.

Camarines Sur National High School Senior High School Scheduling System

According to Marvin Marquez et al. (2022) School class schedule preparation is one of the most important tasks, especially before classes begin. A well-prepared plan will determine whether sections and teachers will get along or if there will be conflicts, so faculty members at the academic institution place a great deal of importance on it. The Camarines Sur National High School (SNHS) is the largest secondary school in the Bicol Region, implementing Senior High School since 2016 with seventy-two (72) sections. The schedule is prepared manually using word processing and spreadsheets, which can lead to conflicts between teachers and students because of the nature of the process itself and the tools being used. This problem was fixed by the CSNHS Senior High School Scheduling System (CSHS-SHSSS), which also lessens the effort of the teachers by automating the schedule preparation of the various CSNHS sections. All instructors and other stakeholders can access this web-based application project by using an online internet connection.

Synthesis

Their study provided the foundation, proving that web-based scheduling can improve efficiency, while our study builds on it by creating a more comprehensive and flexible scheduling and assignment system for ACLC College of Taytay.

Development of a Computerized Faculty Loading, Room Utilization, Subject and Student Scheduling System for Bulacan Polytechnic College

According to Guirre and De Alba (2023), their study aimed to convert the manual scheduling process of Bulacan Polytechnic College into an automated system that handles faculty loading, room utilization, subject timetables, and student scheduling. The paper identifies typical problems in the manual system—confusion, overlapping, wrong room assignments (causing students to move between rooms), and inefficiencies in resource use. They developed a software called AFLRUS4, built with .NET technologies and MySQL, that inputs faculty, courses, sections, and rooms, and computes schedules while detecting conflicts automatically. Through expert validation and user testing (75 respondents), the project was rated “highly acceptable” in functionalities like reliability, efficiency, portability, and maintainability. Their system also helped reduce the time and manpower needed to prepare schedules.

Synthesis

This study supports our project by showing how automated scheduling can integrate teachers, rooms, subjects, and students in one system. It highlights managing multiple resources, not just teacher assignments, and provides useful ideas on usability, evaluation, and conflict detection for building a reliable system.

Foreign Studies

Optimizing the Scheduling of Teaching Activities in a Faculty

According to Diallo and Tudose (2024), their study focused on optimizing the scheduling of teaching activities in a faculty setting through the use of web-based technologies and optimization algorithms. The researchers aimed to improve the efficiency and fairness of class scheduling by developing a system that automatically generates timetables based on various constraints and preferences. Their proposed system was built with an Angular front-end and a Java/Spring Boot back-end, using the Time fold Solve ran intelligent constraint-solving engine that applies algorithms such as Tabu Search and Hill Climbing to produce optimal or near-optimal schedules. The system was designed to consider different types of constraints: hard constraints, which must not be violated (such as preventing class overlaps or ensuring room capacity is not exceeded); medium constraints, which deal with instructor availability and balanced workloads; and soft constraints, which aim to satisfy user preferences such as specific class times or preferred rooms. Through multiple computational experiments, the researchers demonstrated that their model could produce efficient, conflict-free schedules and adapt well to larger datasets. The results also showed improved resource utilization and fairer distribution of teaching loads among faculty members.

Synthesis

Their Study highlights the importance of balancing constraints and user preferences, an approach that can inspire the design and logic of our own web-based scheduling and assignment system. Our study can build on those gaps by including richer modeling of student/teacher mode preferences, and enabling dynamic updates or user edits in real time.

E-SIP: Website-Based Scheduling Information System to Increase the Effectivity of Lecturer's Performance and Learning Process

According to Rahmawati, Sutopo, and Santoso (2021), their study introduced the E-SIP, a website-based scheduling information system designed to enhance the efficiency of academic scheduling in IAIN Tulungagung, Indonesia. The researchers developed the system to solve common issues in manual scheduling, such as overlapping classes, time conflicts, and inefficient use of resources. E-SIP was created using the Waterfall model of software development, which includes the stages of analysis, design, implementation, testing, and maintenance. The E-SIP system allowed administrators to input and manage lecturer schedules, room assignments, and course timetables in a centralized web platform. It automatically generated schedules based on lecturer availability and classroom capacity, minimizing human error and scheduling conflicts. Through testing, the researchers found that the system improved the accuracy and speed of schedule generation, allowing lecturers to focus more on teaching rather than administrative concerns. The study concluded that the system enhanced both lecturer performance and the overall learning process by ensuring more organized and efficient scheduling.

Synthesis

Their research highlights the importance of automation and centralized data management in reducing scheduling conflicts and increasing staff productivity. Our study builds upon these identified gaps by developing a more comprehensive and adaptive web-based scheduling and assignment system. Unlike E-SIP, our system integrates room assignments, teacher schedules, and student section management into a single platform.

Effects of Block Scheduling vs Traditional Period Scheduling on the Academic Achievement of Middle School Students

According to Katelyn Olsen et al. (2020) Students' academic success in middle school is significantly influenced by the way the school is set up. The way kids move from elementary school to high school around the world is greatly influenced by scheduling systems like block scheduling and regular period scheduling. Although there are many factors that contribute to middle school kids' academic success, this senior capstone research project will examine the effects of block scheduling versus traditional period scheduling on middle school students' academic performance. The goal of this project is to determine the optimal middle school scheduling system based on the opinions of administrators, instructors, and students. It also aims to provide ideas for how to best accommodate middle school kids in order to optimize their learning.

Synthesis

This study emphasizes how different scheduling systems, such as block scheduling and traditional period scheduling, can directly influence middle school students' academic achievement. For studies like ours on web-based scheduling systems, these insights highlight the importance of creating flexible and optimized schedules that meet both institutional needs and student learning requirements, ensuring that scheduling solutions contribute meaningfully to educational success.

Development of a Personalized Course Timetable Scheduling System

According to Benjamin Osaze et al. (2020) In order to optimize the assignment of courses to their respective venues based on the number of registered students and venue capacity within the permitted meeting hours, the goal of this study is to create and implement an automated personalized course scheduling system. This was accomplished by gathering information from the teaching staff about their varied system and user requirements, developing the scheduling algorithm's goal function and constraints, defining the system architecture, and putting a working prototype system into place. To gather user and system needs, structured interviews were held with department schedule representatives.

Synthesis

This study provides valuable insights for our web-based scheduling system by showing how personalized, automated scheduling can improve course assignments while considering student numbers, classroom capacity, and instructor preferences. Our project builds on this approach by creating a web-based platform that not only automates scheduling but also makes it easily accessible and interactive, ensuring that administrators, teachers, and students can efficiently manage and view timetables in real time.

Smart Scheduling (SMASCH): Multi-Appointment Scheduling System for Longitudinal Clinical Research Studies

According to Vega et al. (2022) developed SMASCH, a web-based multi-appointment scheduling system that efficiently manages the complex scheduling needs of longitudinal clinical research studies. By automating follow-up visits, optimizing resource allocation, and ensuring data privacy compliance, SMASCH reduces administrative burden and supports participant retention. Its modular, user-friendly design demonstrates how intelligent scheduling algorithms combined with flexible web platforms can handle complex workflows, providing valuable insights for developing web-based systems in other contexts, such as efficiently managing classrooms, teachers, and student groups in educational institutions.

Synthesis

SMASCH supports our study by providing a proven example of how web-based automation and smart scheduling principles can be effectively applied to another field education making scheduling tasks faster, more accurate, and more manageable for administrators and users alike.

Local Studies

Web-Based Faculty Loading and Class Scheduling Management System for ACLC College of Taytay

According to Valenzuela J.P et al. (2024), this study focused on the development of a Web-Based Faculty Loading and Class Scheduling Management System for ACLC College of Taytay to address the difficulties schools face in efficiently managing faculty assignments and class schedules. The system enables administrators to input and manage essential scheduling data such as teacher names, sections, subjects, rooms, and time slots while automatically calculating teaching hours and load distributions. Developed using PHP, JavaScript, Tailwind CSS, Laragon, and Visual Studio, the system provides essential features like printable schedules for easier distribution and record-keeping. It was evaluated using User Acceptance Testing and the ISO 25010 quality standard, focusing on functionality, reliability, usability, security, compatibility, and maintainability. The results showed that while the system performed well overall, there were areas identified for enhancement. The researchers recommended improvements to increase system efficiency, performance, and user-friendliness to better support academic and administrative operations.

Synthesis

This study directly supports the development of our Web-Based Scheduling System by demonstrating how automation and digital management can improve the efficiency of scheduling processes within an academic institution. Like our project, it focuses on creating a user-friendly platform that allows administrators to easily assign teachers, rooms, and time slots while minimizing conflicts and manual errors. The study's use of web technologies such as PHP, MySQL, and JavaScript aligns with our own system's approach, showing the practicality and effectiveness of web-based solutions in managing dynamic scheduling data.

Multi-agent class timetabling for higher educational institutions using Prometheus platform

According to Angelita Guia et al. (2021) Class scheduling is essential to the teaching and learning process in a university setting. Schedules are an essential part of the daily operations of academic institutions. A multi-agent systems-based approach to the university course schedule problem may improve the freedom of each department's class schedule, adaptability in a distributed environment and avoid conflicts between events or resources, and unexpected allocation through agent intervention in a dispersed environment. Most higher education institutions use manual class timing, which is an extremely difficult and time-consuming approach. Building a multi-agent class timing system that automates higher education institutions' (HEIs') class scheduling procedure is the study's primary goal. The Prometheus approach, a comprehensive system covering all stages of software development, has been produced by using the Prometheus technique for the creation of a multi-agent framework implemented in the agent systems.

Synthesis

This study helps our web-based scheduling system by providing ideas on how automation can effectively manage class timetables and reduce scheduling conflicts. While their research uses the Prometheus platform for multi-agent systems, the principles behind their work such as improving efficiency, adaptability, and conflict resolution can guide our system’s development. Their study shows that intelligent scheduling approaches can make class management faster and more organized, which supports our goal of creating a user-friendly, automated web-based scheduling system that enhances accuracy and simplifies administrative work in educational institutions.

Examination Scheduling System for Mapua University

According to Bleyn Christian O. Servaz et al. (2022) In many academic institutions, scheduling and timetabling exams is a difficult completing that necessitates thorough research. Using human scheduling knowledge and experience to create the best possible exam schedule. Without the support of computers, the completion of this operation can be difficult, lengthy, and error-prone due to the state of modern technology. The Mapua University School of Information Technology (SOIT) department had a technical issue with due to their reliance on human techniques and computerized scheduling tools, their examination scheduling procedure with a set output that is exclusive to the class of 2015. This study suggests a web-based mechanism for scheduling exams as a substitute for the technology and exam schedule now used by the school department. It is going to include the idea of computer-based scheduling techniques that are both automated and interactive, and are intended to have the capacity to adapt and produce ideal timetables depending on the dynamic scheduling variables and given limitations on schedule. Additionally, the suggested method has been successfully tested and proved to evaluated by the main participant in the examination scheduling procedure of the designated educational department, the Assistant Technical to the Proctors and the Dean. As a result, acknowledging and embracing it as a substitute for their current procedures for arranging exams.

Synthesis

Their study highlights the benefits of using computer-based solutions that can generate accurate and flexible timetables while meeting institutional constraints. This aligns with our goal of creating a system that allows administrators to easily manage class schedules, assign rooms, and monitor faculty loads. By applying similar automation principles, our study continues the effort to improve scheduling efficiency and support better academic management through technology.

Web Based Course Scheduling System Using Greedy Algorithm

According to John Benedict C Legaspi et al. (2020) The goal of this project was to create a web-based course scheduling system at the FEU Institute of Technology's College of Computer Studies in Manila, Philippines. The developed system uses the Greedy Algorithm to assign assignments to faculty members and manage course scheduling. This study used research and development (R&D) as its methodology. The researcher also talks about the scheduling procedure, which is related to the system's Scheduling Module and uses the Greedy Algorithm as its guiding concept. The system has been put through testing and assessment utilizing FURPS (functionality, usability, reliability, portability, and supportability) criteria.

Synthesis

In summary, applying a web-based course scheduling system with the features and procedures described contributed to a more effective, organized, and technologically advanced scheduling process at ACC College of Taytay. The study's findings and forms provided useful insights for ACC College to enhance its course scheduling system. Researchers also benefited from the design's perceptivity, which served as inspiration and guidance for their own efforts to develop web-based scheduling systems.

Class Schedule: A web-based application for school class scheduling with real-time lazy loading

According to Nuengwong Tuaycharoen et al. (2020) These days, class scheduling poses a serious challenge to educational institutions, instructors, and staff members tasked with creating lesson plans. The more resources that need to be managed-teachers, students, subjects, classrooms, and periods-the more complex the situation becomes. To arrange classes together, numerous department representatives also need to be present at the same location at the same time. To address this issue, we have created a class schedule management system called Class Schedule. Our previous version allowed for Simultaneous multi-user class scheduling, period locking for a period with several teachers and student groups, team teaching, and the effective real-time detection of resource collisions. With our software architecture and loading mechanism (called Lazy Loading), all functions may function in real-time. According to the experimental results, a web page can respond 50% faster than it would take to load the complete page on average when using modern software architecture and lazy loading. Lazy loading can also cut the response time of most pages by 50%.

Synthesis

The study by Nuengwong Tuaycharoen et al. (2020) helps our web-based scheduling system by showing how efficient web technologies can improve class scheduling performance. Their system, Class Schedule, uses real-time features and lazy loading to speed up responses, prevent scheduling conflicts, and support multiple users simultaneously. While our system does not use lazy loading, their approach highlights the value of fast, interactive, and automated scheduling tools. This supports our goal of developing a responsive and efficient web-based scheduling system that ensures accuracy, accessibility, and ease of use for administrators and faculty.

Technical Background

This section presents the various technologies and tools utilized in developing the proposed Web-Based Scheduling and Assignment System for Rooms, Teachers, and Sections at ACLC College of Taytay. Each technology plays a crucial role in ensuring that the system functions efficiently, securely, and accurately. The tools and technologies are categorized into three main components front-end, back-end, and database which work together seamlessly to deliver a functional, responsive, and user-friendly web-based system.

Table 2.1 Technical Background

FRONTEND	BACKEND	DATABASE
HTML	PHP	MySQL
Tailwind CSS	Laravel	
JavaScript	Blade	

Table 2.1 shows the technologies used in developing the Web-Based Scheduling and Assignment System. For the front-end, HTML, Tailwind CSS, and JavaScript were utilized to design a responsive and interactive user interface. The back end used PHP and the Laravel framework to handle system logic and data processing. Meanwhile, Laragon was used as the local development environment to run the web server, manage the database, and test the system during the development process.

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Conceptual Framework

Table 2.2 Conceptual Framework

Software Requirements • HTML • PHP • TAILWIND CSS • JAVA SCRIPT • VISUAL STUDIO • LARAGON • LARAVEL **Hardware Requirements** • LAPTOP • DESKTOP **Peoples Requirements** • ADMIN • TEACHERS • STUDENTS	**Requirements** • Problem/Solutions **Design** • Create system layout and database structure **Development** • Coding/Code review **Testing** • Conduct user testing and feedback **Deployment** • Install and implement the system in the school **Review** • Evaluate system performance and user satisfaction • Gather feedback for improvement	**A Web-Based Scheduling and Assignment System for Rooms, Teachers, and Sections at ACLC College of Taytay**

Based on Table 2.2, our conceptual framework illustrates how we developed the Web-Based Scheduling and Assignment System for Rooms, Teachers, and

Sections at ACLC College of Taytay using the Agile Development Model. In the input phase, we identified the essential requirements needed for system development, which include software, hardware, and people. We used software tools such as HTML, PHP, Tailwind CSS, JavaScript, Visual Studio, Laragon, and Laravel to build and run the system, while desktops and laptops served as our main hardware for development and testing. The people involved in the process—administrators, teachers, and students—helped us by providing feedback and insights that guided our system improvements.

Following the Agile methodology, we went through six iterative phases: Requirements, Design, Development, Testing, Deployment, and Review. Each phase allowed us to refine the system based on user feedback and testing results. Through this process, we were able to create a functional web-based scheduling and assignment system that helps the institution manage class schedules, teacher assignments, and room allocations efficiently. Our goal was to minimize scheduling conflicts, reduce manual errors, and improve the overall efficiency of scheduling operations at ACLC College of Taytay.

CHAPTER III

Methodology

This study employed a Developmental Research Design, which focused on creating and refining a system to address a specific problem. The researchers developed a Web-Based Scheduling and Assignment System to resolve the scheduling challenges at ACLC College of Taytay. To ensure flexibility and continuous improvement, the Agile Software Development Life Cycle (Agile SDLC) methodology was applied. This iterative model allowed the system to be developed in short cycles, where phases such as planning, designing, developing, and testing were repeatedly refined based on user feedback from teachers, students, and administrators. The use of Agile SDLC promoted collaboration, adaptability, and a faster response to changes, ensuring that the final system effectively met user needs.

Research Design

This study used a Quantitative Research Design to assess the effectiveness of the Web-Based Scheduling and Assignment System for Rooms, Teachers, and Sections at ACLC College of Taytay. The quantitative approach focused on collecting measurable data through survey questionnaires answered by administrators, IT experts, Teachers, and Students.

System Development Life Cycle

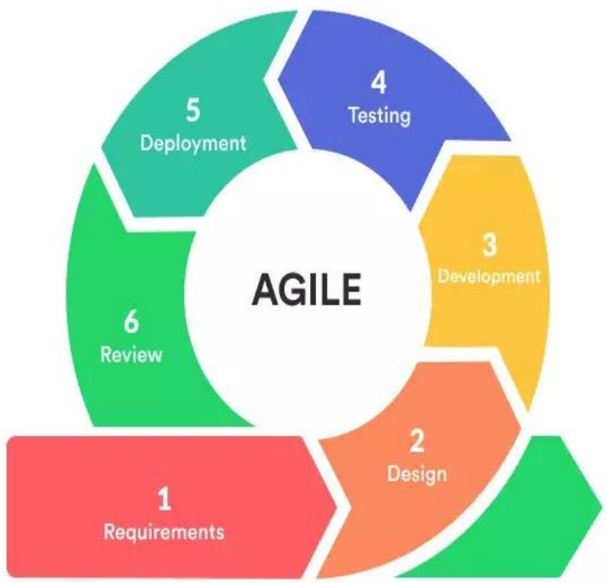

Figure 3.1 Development Model

This study adopted the Agile Development Model as the framework for developing the Web-Based Scheduling and Assignment System for Rooms, Teachers, and Sections at ACLC College of Taytay. The Agile model was chosen because it promotes flexibility, collaboration, and continuous improvement throughout the system development process. It allowed the researchers to respond efficiently to changes and feedback from users during each phase of development. The Agile methodology divided the development process into multiple iterations or sprints, with each sprint producing a functional version of the system. This iterative approach ensured that the system was continuously tested, refined, and improved based on user input and evaluation.

System Development Procedures

Requirements Phase

In this phase, data were gathered regarding the current manual scheduling process of the school. Problems such as scheduling conflicts, time consumption, and difficulty in managing faculty loads were identified. System requirements were defined based on the needs of administrators and users.

Design Phase

The system design was planned, including the database structure, system architecture, and user interface. The researchers designed the layout of the admin dashboard, scheduling module, and data input forms. The workflow of the system—from data input to schedule generation was also established.

Development Phase

The actual coding of the system was conducted using PHP. Core features such as automatic scheduling, room assignment, section management, and faculty load computation were implemented. The admin dashboard was also developed to manage all system operations.

Testing Phase

The system was tested to ensure that all functions were working correctly. Errors such as duplicate schedules, incorrect assignments, and system bugs were identified and fixed. User testing was also conducted to evaluate the system's functionality and usability.

Deployment Phase

After successful testing, the system was deployed and implemented in the school environment. The system was published on hosting, and administrators were trained on how to use it effectively for scheduling tasks.

Review Phase

Feedback was collected from users after deployment. The system was evaluated based on performance, usability, and efficiency. Necessary improvements and updates were identified and applied in the next development cycle.

Implementation Phase

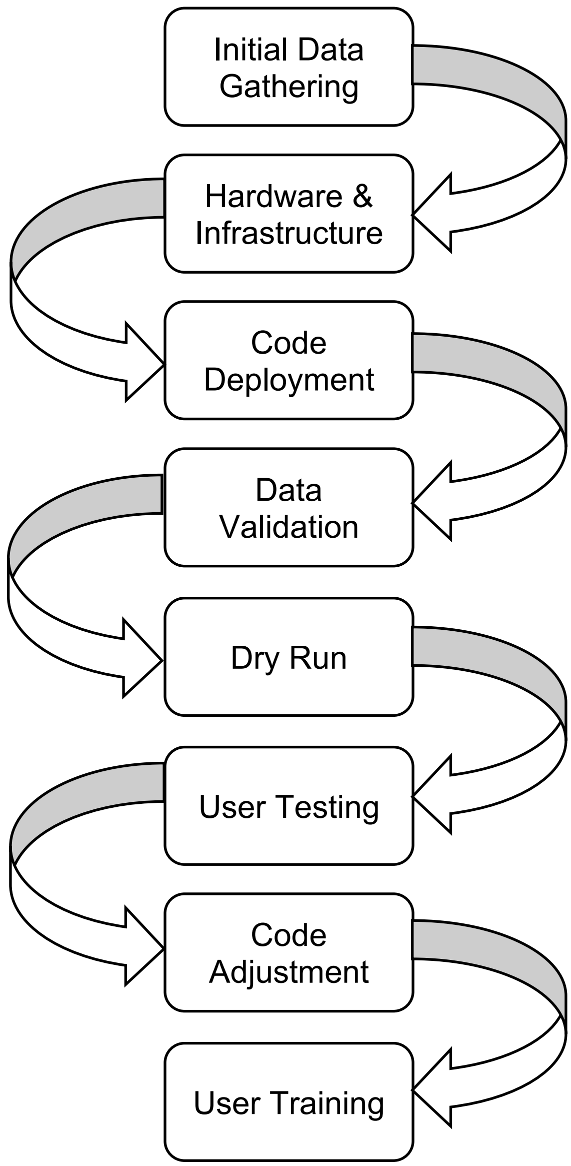

Figure 3.2 Implementation Phase

Initial Data Gathering

We Researchers gathered initial data from ACLC College of Taytay.

Hardware/Infrastructure

We researchers checked if ACLC College of Taytay had computer units and an internet connection to access the system.

Code Deployment

We researchers handed over the system to ACLC College of Taytay after it was completed.

Data Validation

We researchers tested the system for data entry errors and bugs after it was completed.

Dry Run

We researchers conducted a dry run using sample scenarios. These scenarios were participated in by the researchers and the management.

User Testing

We system was deployed for client testing after the researchers completed the dry run and confirmed that the system was working.

Code Adjustment

We researchers conducted code adjustments whenever errors or bugs were found from the recent testing.

User Training

Following the system's implementation at ACLC College of Taytay, the users were trained on how to use the system. The researchers educated the administrators about the system's features and instructed them on how to handle the status of appointments. The researchers also showed the administrators how to keep the website's information up to date.

Testing Phase

The testing phase ensured that the system performed correctly and met user requirements. The researchers conducted User Acceptance Testing (UAT) to evaluate usability, reliability, and performance. The system was also assessed using the ISO 25010 software quality model to measure key factors such as functionality, security, maintainability, and efficiency. Feedback from users was gathered and used to fix errors, improve design, and enhance performance. This phase guaranteed that the system was ready for deployment and capable of providing accurate, conflict-free schedules for ACLC College of Taytay.

Tools, Technologies, and Techniques

This section presented the different tools, technologies, and techniques used in the development of the proposed Web-Based Scheduling and Assignment System for Rooms, Teachers, and Sections at ACLC College of Taytay. These components were carefully selected to ensure that the system was efficient, secure, and easy to use. Each tool and technology served a specific purpose in the design, development, testing, and maintenance of the system. The following tables showed the APIs, tools, frameworks, libraries, and applications used throughout the development process.

Table 3.1 Tools, Technologies and Technique

Application	Tools	Framework	Libraries	API
Microsoft Word	Composer	Laravel 12	Tailwind CSS	Fetch API
Canva	NPM	Blade	Axios	Laravel AJAX/JSON endpoints
Draw.io	Vite		Alpine.js	PDO (MySQL)
Microsoft Excel	Laravel Artisan			

Table 3.1 showed the tools, technologies, and techniques used in the development of the Web-Based Scheduling and Assignment System. Applications such as Microsoft Word, Canva, Draw.io, and Microsoft Excel were used for documentation, system design, diagram creation, and data organization. Development tools such as Composer, NPM, Vite, and Laravel Artisan were

utilized to manage dependencies, build processes, and streamline the development workflow.

The system was developed using Laravel 12 as the primary framework for backend development, while Blade was used as the templating engine for structuring the user interface. Tailwind CSS was applied to design a responsive and modern layout, and Alpine.js was used to add lightweight interactivity to the frontend. Axios was implemented as a library for handling HTTP requests and enabling efficient communication between the frontend and backend systems. For data exchange and integration, the Fetch API and Laravel-based AJAX/JSON endpoints were used to support dynamic and real-time interactions within the system.

Software Requirements

The Web-Based Scheduling System for ACLC College of Taytay used a PC with a web browser, Laragon, Visual Studio Code, and MySQL Server, running on Windows 7 or higher. The system was designed to be useful, efficient, and simple to operate.

Hardware Requirements

The system used a PC or laptop with reasonable specifications to be able to code and develop the system.

Table 3.2 Hardware

Hardware	Minimum	Recommended
Processor	Intel i3 and equivalent to another brand	Intel Core i5 or higher
Memory (RAM)	4GB	8GB or higher
Storage	256GB HDD	512GB SSD or higher
Internet Connection	LAN/Wi-Fi Capability	Gigabit Ethernet/ Stable High-speed Internet
Display	1366x768 resolution	1920x1080

Table 3.2 showed the minimum and recommended hardware requirements for developing and running the Web-Based Scheduling and Assignment System. The system required at least an Intel i3 processor, 4GB of RAM, and 256GB of storage

to operate properly, while higher specifications such as an Intel i5 processor, 8GB of RAM, and 512GB SSD were recommended for better performance. A stable internet connection and a display resolution of at least 1366×768 were needed, with a high-speed connection and 1920×1080 resolution preferred for smoother and clearer system operation.

Data Gathering Procedure

To gather statistical data, the researchers used purposive sampling, selecting participants such as students, administrators, and teachers who were directly involved in the scheduling process. The questionnaire was reviewed by a grammarian to ensure clarity and correctness, while statisticians assisted in refining the methods, developing appropriate analyses, and interpreting the collected data to ensure accurate and reliable results.

Data Gathering Instrument

Gathering data was a basic requirement for research. The requirements for the creation of the proposed Web-Based Scheduling and Assignment System for Rooms, Teachers, and Sections at ACLC College of Taytay were identified through thorough investigation and data collection. This process was an essential step in analyzing, designing, and successfully developing the proposed system.

We employed a variety of techniques to obtain factual information from the respondents of the study. The primary instrument used to collect data was a standard questionnaire based on the ISO 25010 software quality model. The questionnaire was designed to assess the quality characteristics that the proposed system met, such as functionality, reliability, usability, efficiency, maintainability, and portability.

Questionnaire - The questionnaire was the main tool used for gathering data. We prepared a form containing statements and questions related to the system's features and performance, aligned with the ISO 25010 quality criteria. The questionnaire was designed to be convenient for both respondents and researchers. Respondents, such as administrators, teachers, and students, were given enough time to answer the form during their free hours to ensure thoughtful and accurate responses. This method ensured that the data gathered were reliable, relevant, and useful for evaluating the effectiveness and quality of the proposed system.

Profile of the Respondents

This was used to compare the respondents' demographic data, including name, gender, and the quantity of samples that the researchers collected for statistical analysis.

Respondents of the Study

The respondents of the study were composed of the administrator, teachers, students, and IT experts of ACLC College of Taytay. These respondents were selected because they represented the key groups who interacted with or evaluated the proposed Web-Based Scheduling and Assignment System for Rooms, Teachers, and Sections.

The administrator served as the main user responsible for managing class schedules, teacher assignments, and room utilization. The teachers represented the users who received and managed their teaching schedules through the system. The students served as end-users who accessed and viewed their class schedules online. The IT experts were included to provide technical evaluation and validation of the system based on software quality standards, ensuring its reliability, functionality, and overall performance.

Table 3.3 Respondents

Respondent	Sample
Administrator	5
Teachers	10
Students	30
IT Experts	5

Table 3.3 showed that the sampling had a total of 50 respondents, including 5 administrators, 10 teachers, 30 students, and 5 IT experts.

SETTING OF THE STUDY

The study was conducted at ACLC College of Taytay. To improve the existing scheduling process of the institution, the proposed Web-Based Scheduling and Assignment System was implemented within the school premises. The system aimed to automate the assignment of rooms, teachers, and sections, making it more efficient and organized.

STATISTICAL TREATMENT OF DATA

The statistical treatment depended on the environment of the research; the researchers used different computations to analyze the gathered data.

1. Mean - We Researchers and Statisticians computed the mean by summing all individual data values and dividing the sum by the total number of data points.

2. Percentage - We Researchers and Statisticians used percentage to determine what portion of the total respondents provided a specific response.

$$P = \frac{N}{T} \times 100$$

Where:

P = Percentage

N = Number of respondents who selected a specific response

T = Total number of respondents

3. Likert Scale – We Researchers and Statisticians were able to gather quantitative data about individuals' subjective experiences or opinions through the utilization of Likert scales. After acquiring responses, the data were analyzed by computing the mean and other statistical measures to determine the distribution and patterns of opinions within the group.

Table 3.4 Likert Scale

Scale	Ratings	Verbal Interpretation
5	4.20 – 5.00	Highly Agree
4	3.40 – 4.19	Agree
3	2.60 – 3.39	Moderately Agree
2	1.80 – 2.59	Disagree
1	1.0 – 1.79	Highly Disagree

4. Frequency - This was used to determine the actual number of respondents and their demographic profile.

Weighted Mean - We researchers and Statisticians used measures of central tendency to determine where the majority of the respondents' answers fell within a question cluster.

$$x = \frac{\Sigma W_n x_n}{\Sigma W_n}$$

Where:

x = Weighted Mean

Σ = Summation (the sum of all values)

W_n = Weight assigned to each variable or response

x_n = Value or score of each item

Standard Deviation - The researchers used standard deviation to determine the degree of variation or dispersion of responses within a group.

$$SD = \sqrt{\frac{\Sigma(x - \overline{X})^2}{n}}$$

Where:

SD = Standard Deviation

Σ = Sum of the squared differences

X = Each individual data point

$\bar{X}$ = Mean of the data

N = Total number of data points

Use Case Diagram

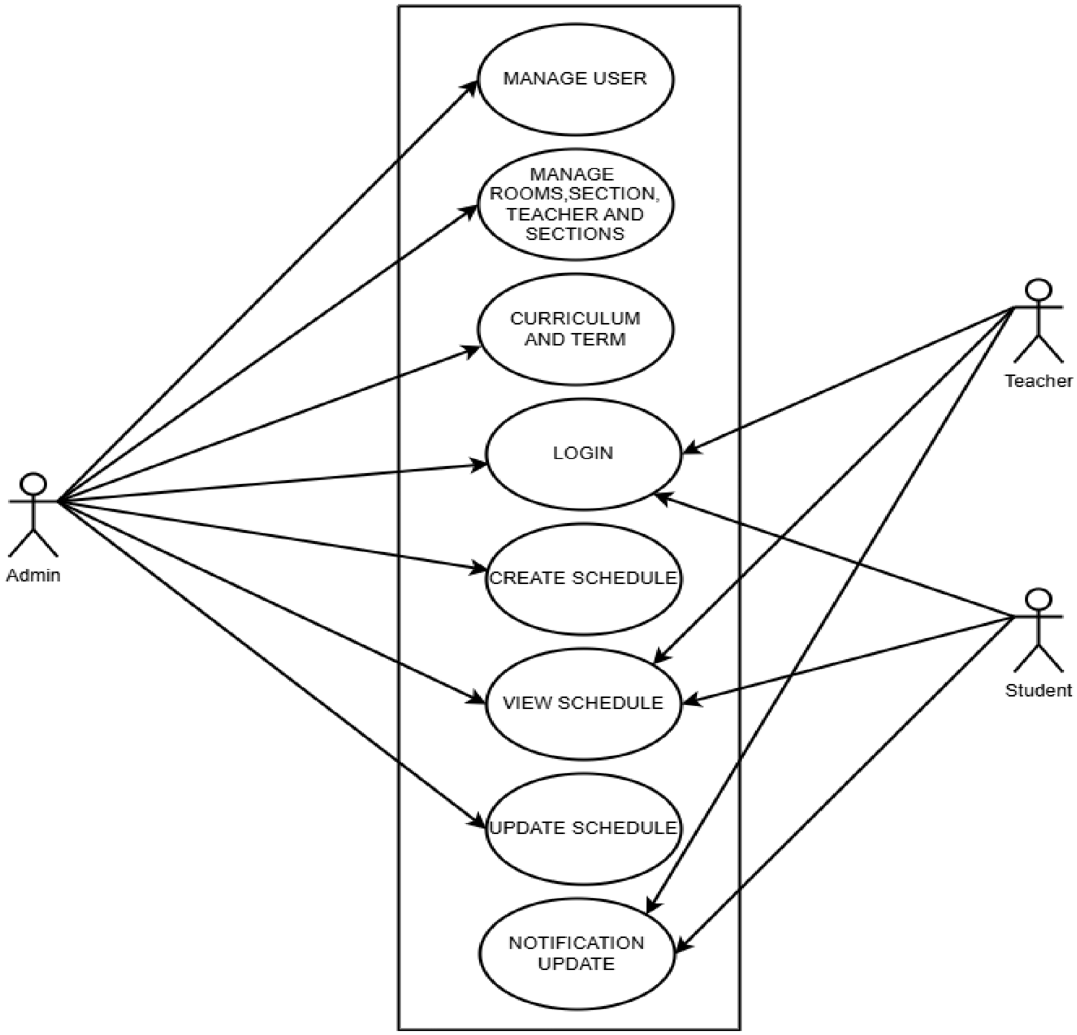

Figure 3.3 Use Case Diagram

In Figure 3.3, the use case diagram illustrated the interaction between the users and the system, showing how the Admin, Teacher, and Student performed different functions. The Admin had full control over the system, allowing them to manage users, rooms, sections, teachers, and curricula, as well as create, view, and update schedules. The Admin could also handle notifications and system updates. On the other hand, both Teachers and Students could log in to the system to view their schedules and receive notifications or updates related to classes or announcements. Overall, the diagram showed that the Admin managed and maintained the system, while Teachers and Students mainly accessed information related to their schedules and notifications.

CONTEXT DIAGRAM

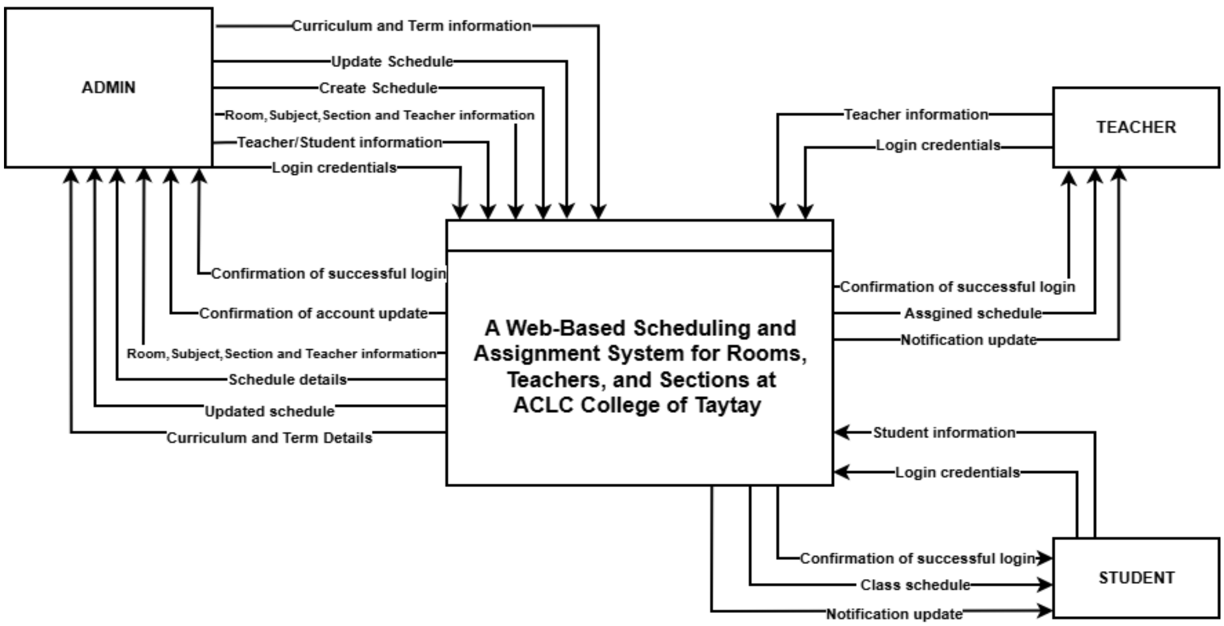

Figure 3.4 Context Diagram

Figure 3.4 showed the Context Diagram of the Web-Based Scheduling and Assignment System for Rooms, Teachers, and Sections at ACLC College of Taytay. It illustrated how the system interacted with its external entities, namely the Admin, Teacher, and Student, by showing the flow of information between them. In this diagram, the Admin was responsible for managing the overall system operations. The Admin inputted data such as room, subject, section, teacher, student information, and curriculum and term details. The Admin could also create and update schedules. In return, the system provided the Admin with confirmations of successful logins, account updates, and schedule updates.

The Teacher interacted with the system by providing login credentials and receiving confirmation of successful login. The system provided the Teacher with their assigned schedule and sent notification updates regarding any changes or announcements.

Similarly, the Student logged in using their credentials and received confirmation of successful login. The system provided the student with their class schedule and notification updates about any class-related information.

Overall, Figure 3.4 presented how the system acted as a central hub that facilitated smooth communication and data exchange between the Admin, Teacher, and Student to ensure efficient scheduling and management within the institution.

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ANALYTICAL TOOLS

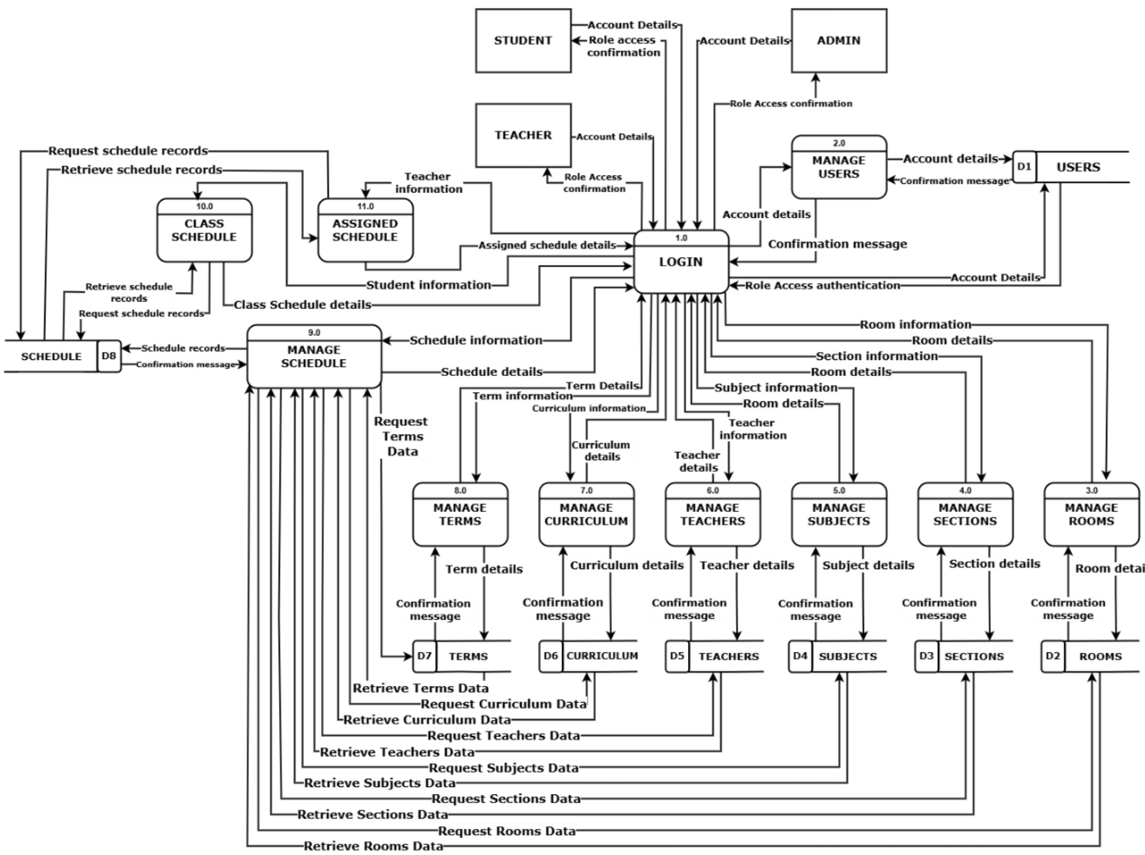

Figure 3.5 Class Scheduling Data Flow Diagram

Figure 3.5 showed the Data Flow Diagram of the Web-Based Scheduling and Assignment System for Rooms, Teachers, and Sections. The diagram illustrated how data moved between the system's main processes, data stores, and external entities such as the administrator, teachers, and students. The system began with user authentication through the login process, which verified credentials and assigned access levels based on roles. The administrator could manage users, rooms, sections, subjects, teachers, curriculum, and terms, all of which were stored in their respective databases. These data were used in the Manage

Schedule process, where the admin created and assigned class schedules while the system automatically checked for conflicts. Teachers could view their assigned schedules, while students could access their class schedules by section.

ENTITY RELATIONSHIP DIAGRAM

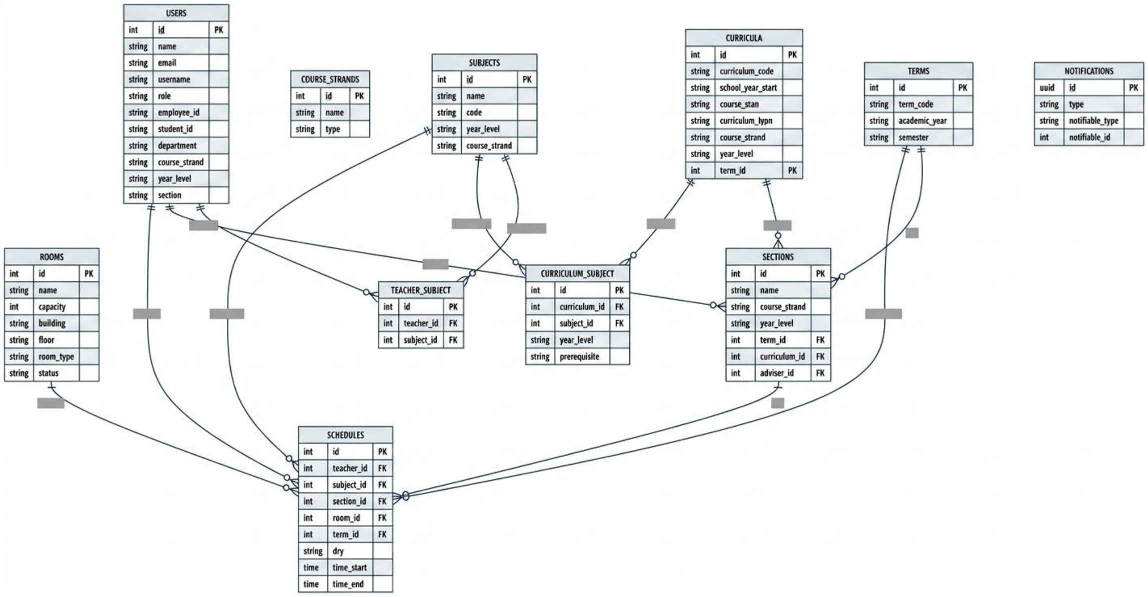

Figure 3.6 Entity Relationship Diagram

In figure 3.6 the Entity Relationship Diagram illustrated the Web-Based Scheduling and Assignment System for Rooms, Teachers, and Sections. It showed the relationship between key entities such as Users, Teachers, Subjects, Sections, Rooms, Curriculum, Terms, and Schedules. The database design ensured that all records were properly linked to support efficient schedule creation and conflict checking. Administrators managed user accounts, rooms, subjects, and terms, while schedules were generated and stored systematically in the

Schedules. This design promoted data consistency, prevented duplication, and maintained the integrity of all academic scheduling information.

User Acceptance Testing

Table 3.5 Sample UAT Test Case

Test Case ID	Test Scenario	Test Steps
TC-01	Verify valid login credentials	1. Go to login page. 2. Enter valid username and password. 3. Click "Login" button.
TC-02	Verify invalid login credentials	1. Enter wrong username/password. 2. Click "Login"
TC-03	Verify account creation by admin	1. Log in as Admin. 2. Go to the “User Management” module. 3. Click “Add User”. Enter User details and click “Save”.
TC-04	Verify room management (add, edit, delete)	1. Log in as Admin. 2. Go to “Room Management”. 3. Click “Add Room” and enter room details. 4. Click “Save” and verify it appears in the list. 5. Edit room information and save again. 6. Delete a room and verify it’s removed from the list.
TC-05	Verify teacher management (add, edit, delete)	1. Log in as Admin. 2. Go to “Teacher Management”. 3. Click “Add Teacher” and input details. 4. Save and confirm teacher appears in the list. 5. Edit teacher data and save changes. 6. Delete teacher record and verify removal.

TC-06	Verify section management (add, edit, delete)	1. Log in as Admin. 2. Go to “Section Management”. 3. Click “Add Section” and enter section details. 4. Save and confirm section appears in the list. 5. Edit section details and save changes. 6. Delete section and verify removal.
TC-07	Verify subject management (add, edit, delete)	1. Log in as Admin. 2. Go to “Subject Management”. 3. Click “Add Subject” and input subject details. 4. Click “Save” and verify subject appears in the list. 5. Edit subject details and update record. 6. Delete subject and confirm it’s removed from the list.
TC-08	Verify curriculum management (add, edit, delete)	1. Log in as Admin. 2. Go to “Curriculum Management”. 3. Click “Add Curriculum” and input subjects’ details. 4. Save and confirm the curriculum appears in the list. 5. Edit curriculum details and click “Update”. 6. Delete a curriculum and verify it’s removed.
TC-09	Verify term management (add, edit, delete)	1. Log in as Admin. 2. Go to “Term Management”. 3. Click “Add Term” and input semester/term details. 4. Save and confirm term appears in the list. 5. Edit term details and click “Update”. 6. Delete a term and confirm it’s removed.
TC-10	Verify schedule creation	1. Log in as Admin. 2. Go to “Schedule Management”.

		3. Select a term, subject, teacher, room, and section. 4. Click “Save”. 5. Verify that the schedule appears in the list.
TC-11	Verify conflict detection in scheduling	1. Log in as Admin. 2. Try assigning the same room or teacher to two different schedules at the same time. 3. Verify that the system prevents saving and displays a conflict warning.
TC-12	Verify teacher viewing assigned schedule	1. Log in as Teacher. 2. Click “My Schedule”. 3. Verify that the correct subjects, rooms, and times are displayed.
TC-13	Verify student viewing class schedule	1. Log in as Student. 2. Click “My Schedule”. 3. Verify that the schedule for the student’s section is displayed correctly.
TC-14	Verify logout functionality	1. Log in as any user. 2. Click “Logout” from the dashboard. 3. Verify that the system redirects to the login page.

Evaluation Procedures

The evaluation procedure was conducted to assess the quality of the Web-Based Scheduling and Assignment System for Rooms, Teachers, and Sections based on the ISO 25010 Software Quality Model. This evaluation was performed by selected IT and technical experts to ensure that the system met professional software standards in terms of functionality, performance, and usability.

The gathered feedback was then analyzed to determine the system's overall quality and effectiveness based on the following ISO 25010 criteria:

Functionality – measured how well the system performed its intended tasks and fulfilled the needs of its users.

Reliability – evaluated the system's stability and consistency in operation, ensuring that it performed accurately without frequent errors or failures.

Usability – focused on how easy and convenient it was for users to navigate and interact with the system's interface.

Efficiency – assessed how effectively the system utilized resources and responded to user inputs, ensuring fast and smooth performance.

Maintainability – referred to the ease with which the system could be modified, updated, or enhanced to adapt to future requirements.

Portability – evaluated the system's ability to operate on different platforms or environments without significant reconfiguration or errors.

CHAPTER IV

Results and Discussions

This chapter presented and discussed the results of the Web-Based Scheduling and Assignment System. The findings were based on the data gathered and were organized according to the objectives of the study. It also explained how the system improved scheduling and assignment processes.

Presentation of Objectives

Presentation of Objective 1 (To Design and Create/Develop)

1.1 Design and develop a secure web-based scheduling system that features user authentication and role-based access for administrators, teachers, and students.

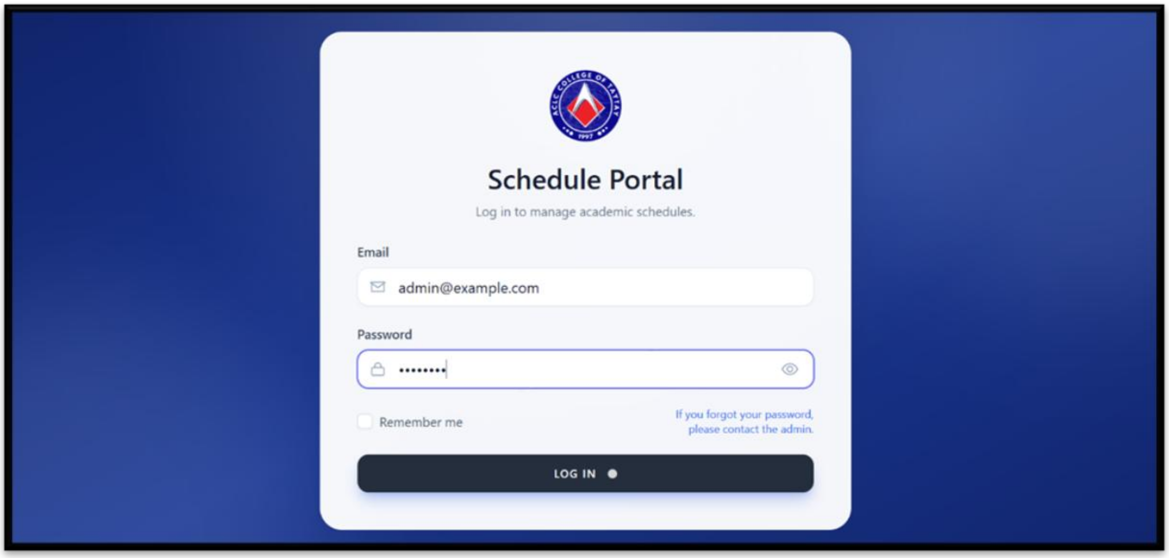

Figure 4.1 Login Page

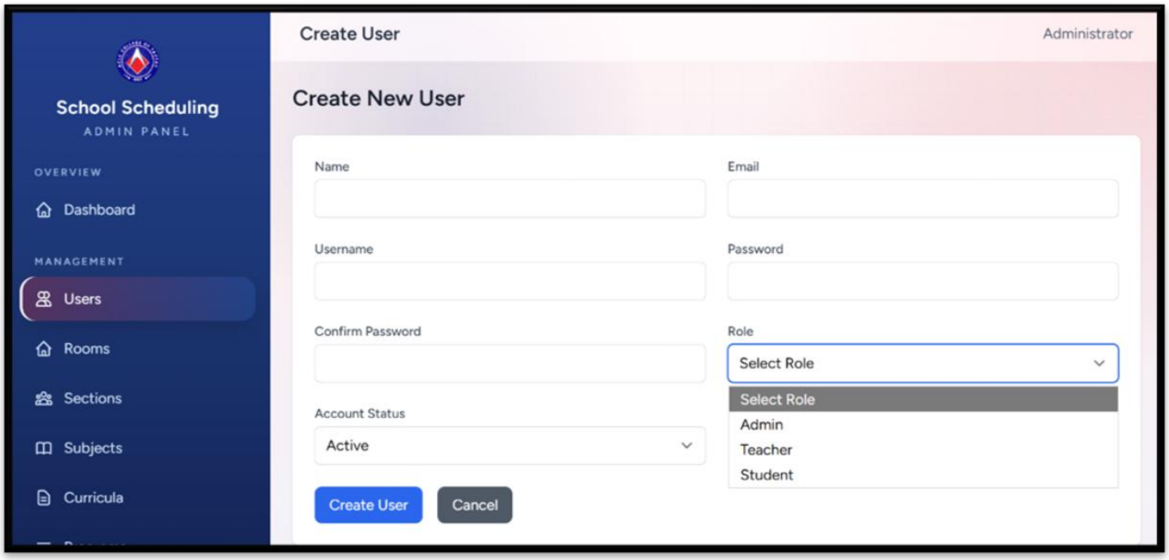

Figure 4.2 Role-based Access Features

Figure 4.1 and 4.2 presented the implementation of the system's user authentication and role-based access features. The system provided a secure login interface that required valid user credentials before granting access. In addition, it included an administrative module that allowed the creation and management of user accounts with designated roles such as administrator, teacher, and student. This functionality ensured controlled access to system features and supported efficient management of users within the scheduling system.

1.2 Provide the administrator with full management control to add, edit, delete, and view records of teachers, students, rooms, subjects, and sections.

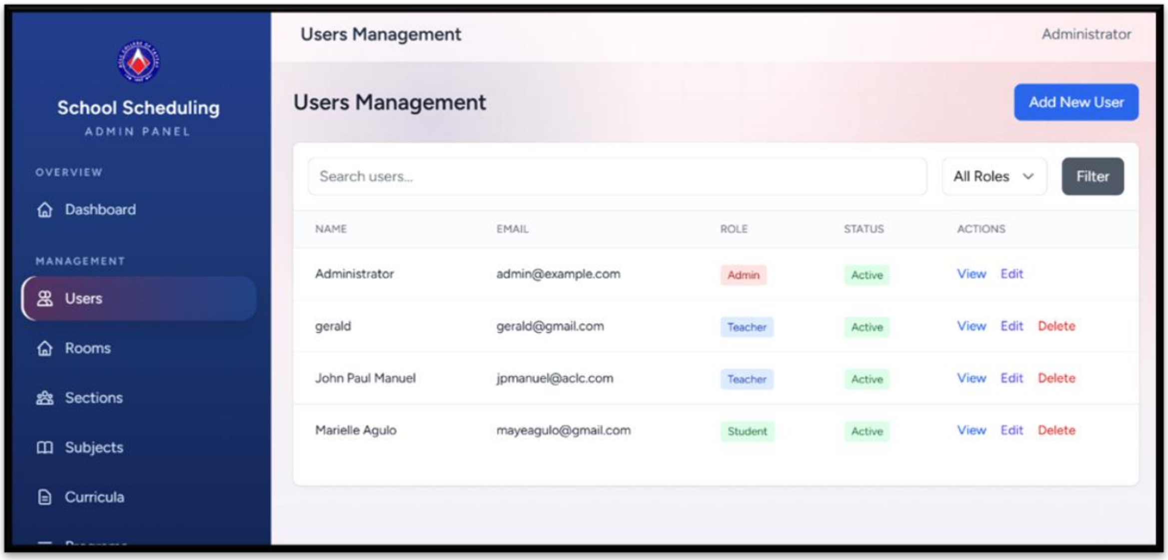

Figure 4.3 Users Management

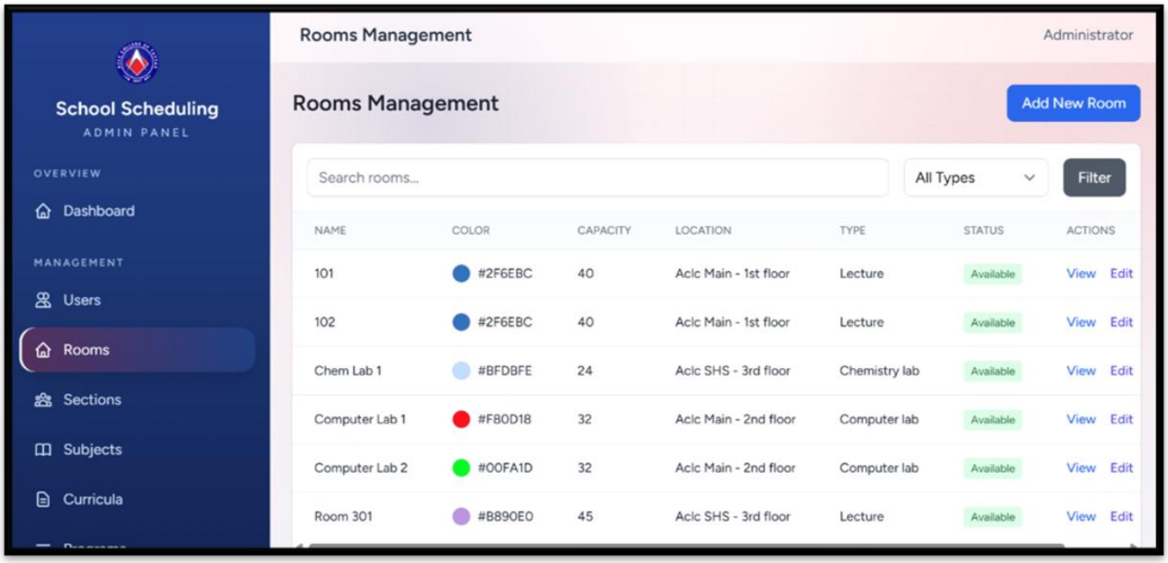

Figure 4.4 Rooms Management

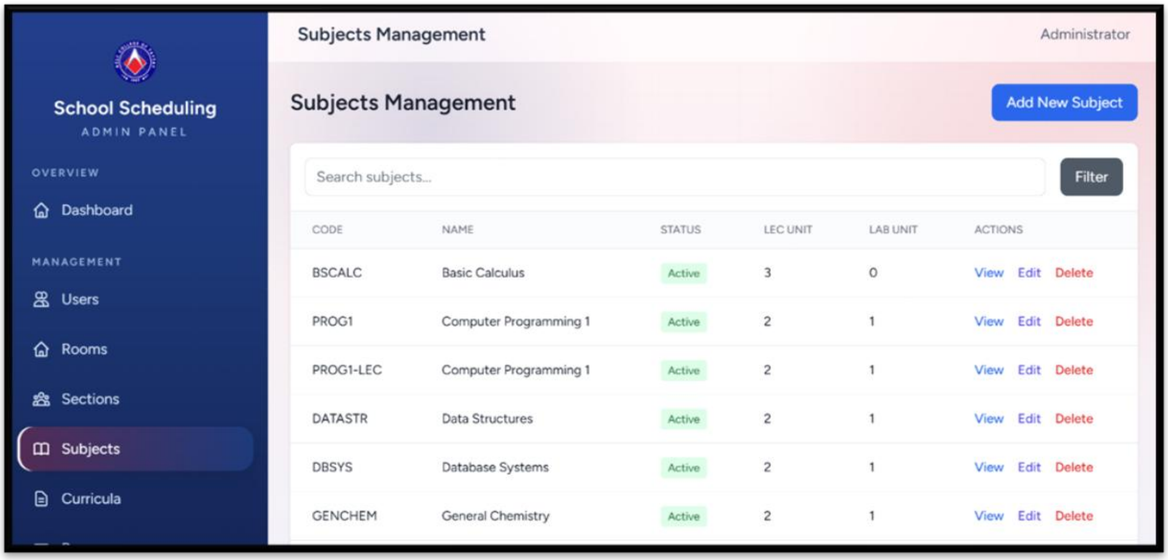

Figure 4.5 Subjects Management

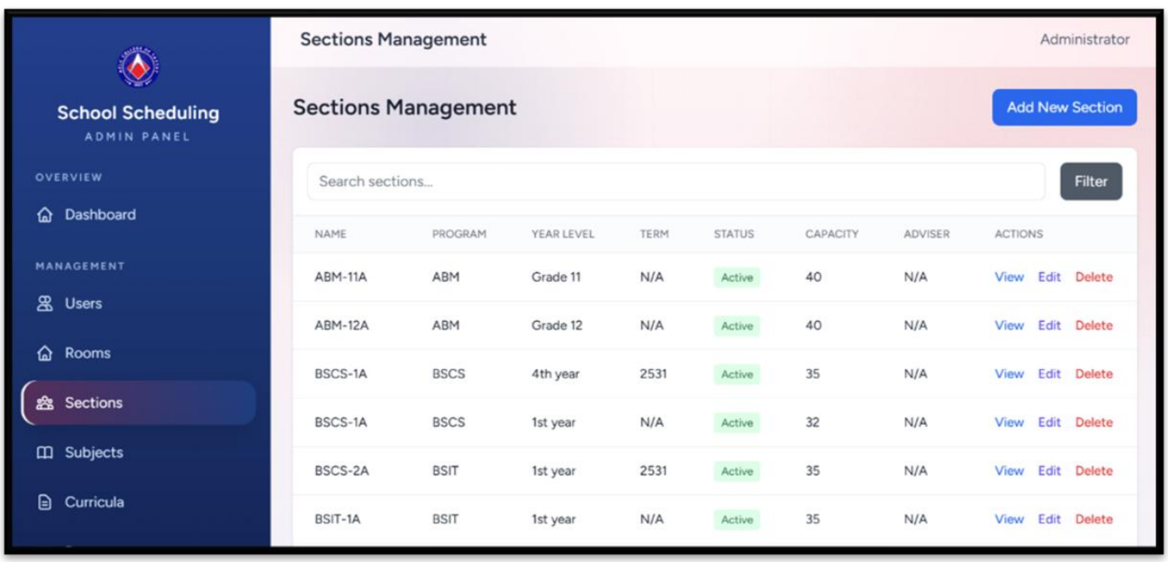

Figure 4.6 Sections Management

Figure 4.3 to 4.6 presented the system's administrative management functions, which provided full control over essential records. The system enabled the administrator to efficiently manage users, rooms, sections, and subjects through features that allowed adding, editing, deleting, and viewing information.

This functionality supported organized data handling and ensured accurate and up-to-date records, contributing to the overall efficiency of the scheduling system.

1.3 Develop a generated scheduling module that allows the administrator to assign teachers, subject, and rooms to class sections while detecting time and room conflicts

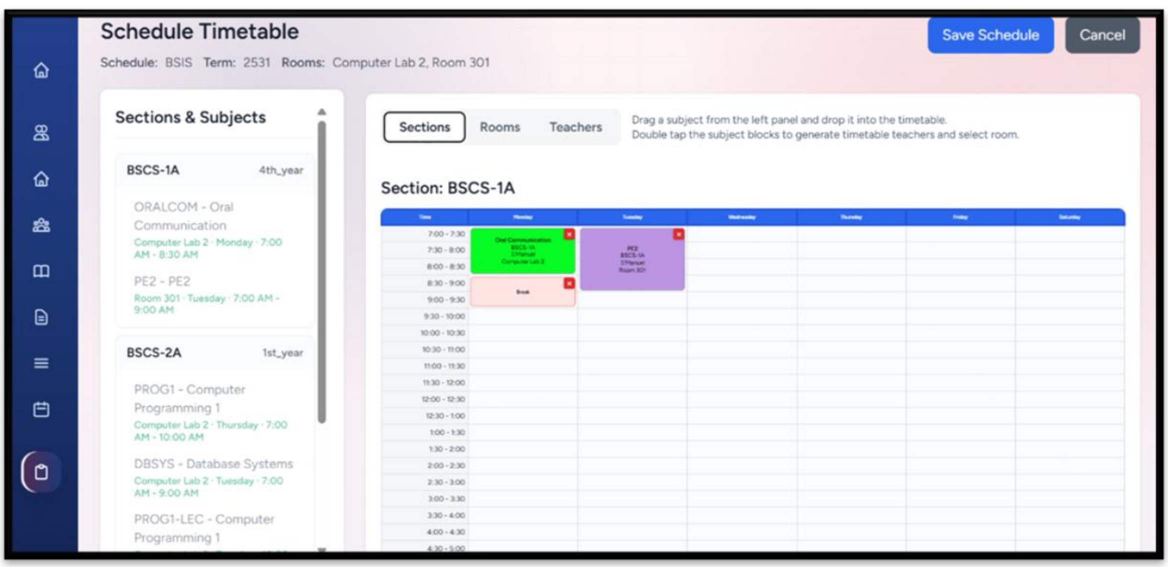

Figure 4.7 Schedule Timetable

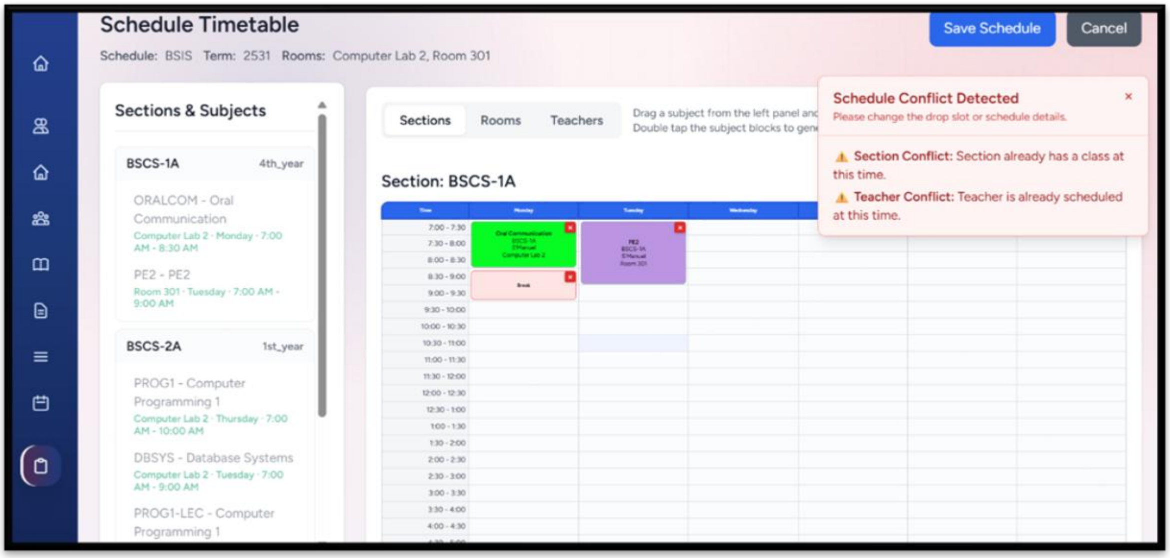

Figure 4.8 Conflict Detection

Figure 4.7 and 4.8 presented the generated scheduling module of the system. This feature enabled the administrator to assign teachers, subjects, and rooms to specific class sections through a structured timetable interface. It also incorporated conflict detection mechanisms that helped prevent overlapping schedules in terms of time and room allocation, ensuring an organized and efficient scheduling process.

1.4 Enable teachers and students to view their respective schedules through personalized "My Timetable" that display subject, time, and room assignments.

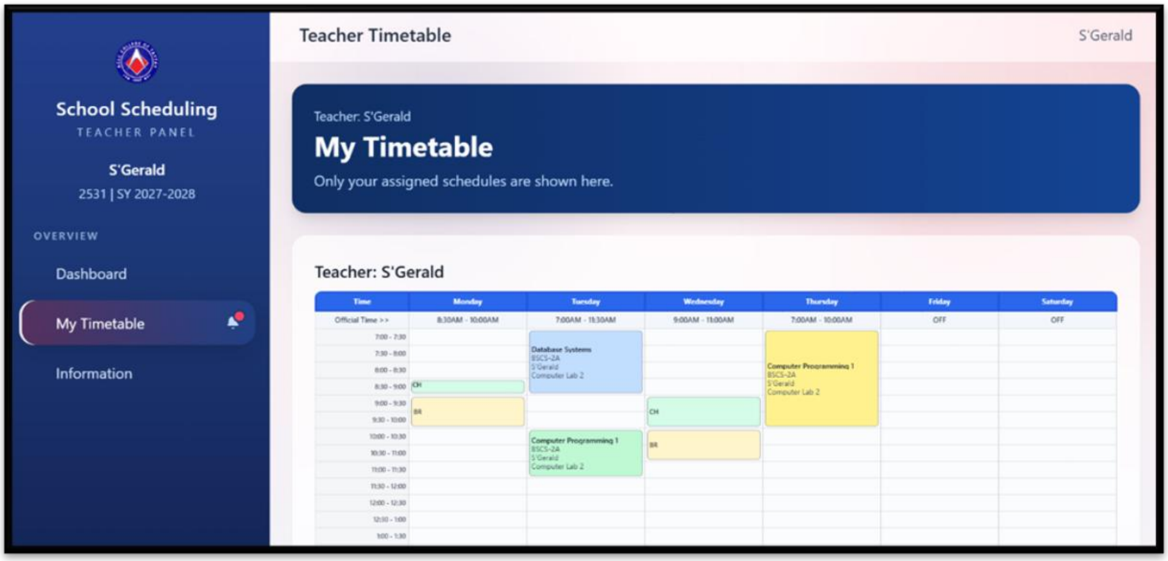

Figure 4.9 Teachers Respective Schedules

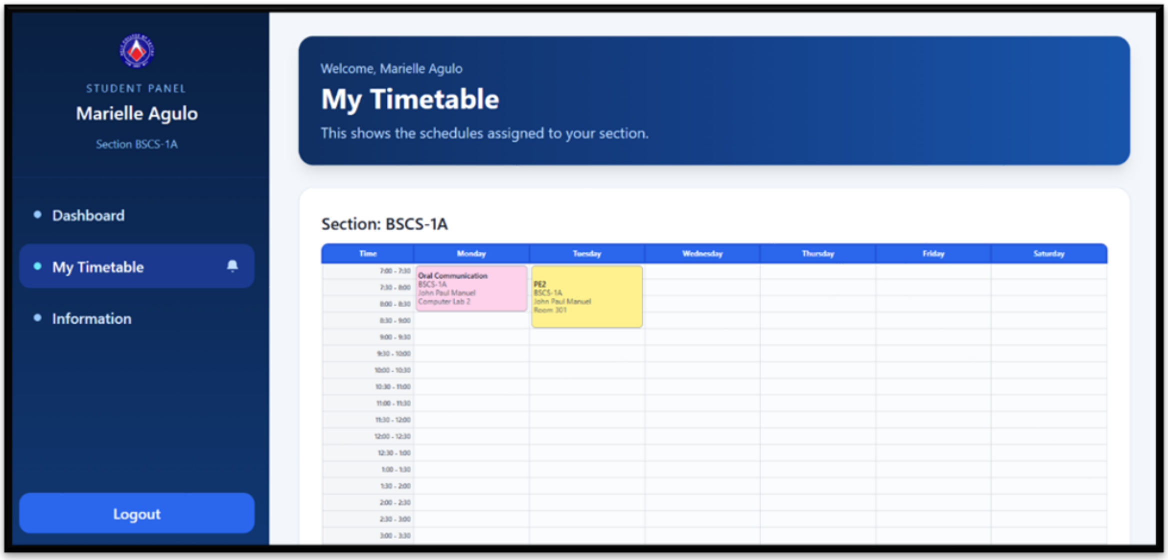

Figure 4.10 Student's Respective Schedules

Figure 4.9 and 4.10 showed the "My Timetable" feature, where teachers and students could view their personalized schedules in a weekly format. It displayed subjects, time, and room assignments using color-coded blocks for easy understanding. Teachers saw only their assigned classes, while students viewed schedules specific to their section, providing a clear and organized way to track daily activities.

1.5 Implement a notification feature to inform users of updates, changes, or new schedules in real time.

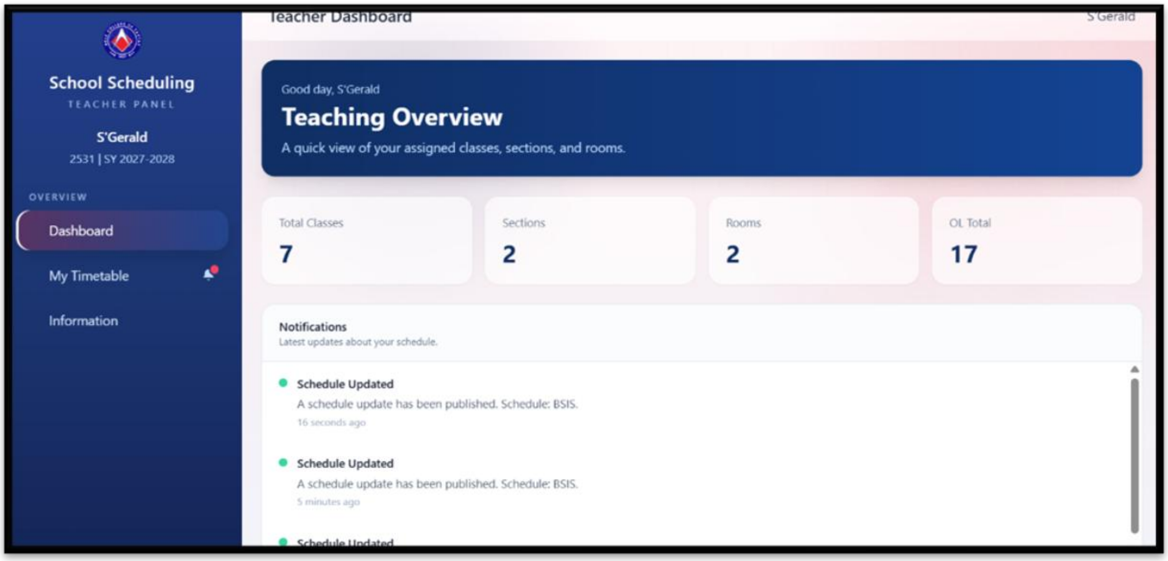

Figure 4.11 Teachers Notification

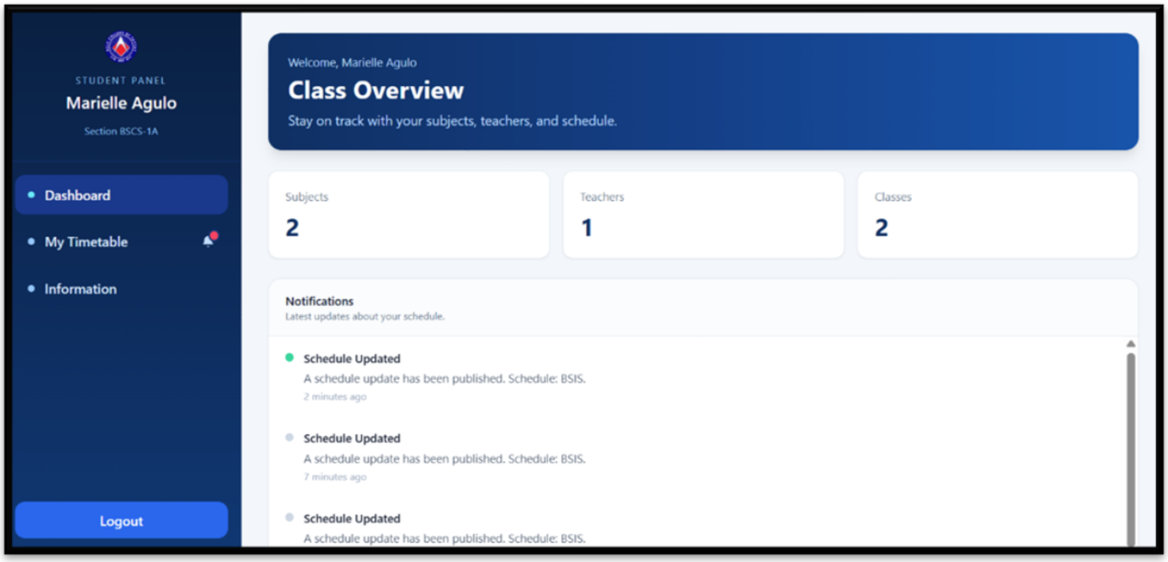

Figure 4.12 Students Notification

Figure 4.11 and 4.12 showed the implementation of a notification feature in the system, where both student and teacher dashboards displayed real-time updates about schedule changes, newly published timetables, and other important announcements.

Presentation of Objective 2 (To Test)

UAT Results

Table 4.1 To Test

Test Case ID	Test Scenario	Expected Result	Actual Result	Status (Pass/Fail)	Remarks
TC-01	Verify valid login credentials	User successfully logs in and is redirected to dashboard	User logged in successfully and dashboard displayed.	Pass	
TC-02	Verify invalid login credentials	System displays error message and prevents login	Error message shown: “Invalid credentials”	Pass	
TC-03	Verify account creation by admin	New user is added and appears in user list	| User successfully added and visible in list	Pass	
TC-04	Verify room management	Room can be added, edited, and deleted successfully	All operations performed	Pass	
TC-05	Verify teacher management	Teacher record can be managed (add/edit/delete)	User logged in successfully and dashboard displayed.	Pass	
TC-06	Verify section management	Section can be added, edited, and deleted	Section successfully managed	Pass	
TC-07	Verify subject management	Subject is added, updated, and deleted properly	Subject management works correctly	Pass	

TC-08	Verify curriculum management	Curriculum can be created, edited, and deleted	All actions completed successfully	Pass	
TC-09	Verify term management	Term is added, updated, and deleted	Term management functioning properly	Pass	
TC-10	Verify schedule creation	Schedule is saved and displayed in list	Schedule successfully created and shown	Pass	
TC-11	Verify conflict detection	System prevents duplicate time/room/teacher conflicts	Conflict warning displayed correctly	Pass	
TC-12	Verify teacher viewing schedule	Teacher sees correct assigned schedule	Schedule displayed accurately	Pass	
TC-13	Verify student viewing schedule	Student sees correct section schedule	Schedule displayed correctly	Pass	
TC-14	Verify logout functionality	User is logged out and redirected to login page	Logout successful and redirected	Pass	

Presentation of Objective 3 (To Evaluate)

The system was evaluated using the ISO 25010 Software Quality Model, which measures software quality in terms of functionality, reliability, usability, performance efficiency, security, compatibility, maintainability, and portability.

Based on the results gathered from administrators, teachers, students, and IT experts, the system received high ratings across all criteria. This indicates that the system is efficient, user-friendly, secure, and capable of handling scheduling tasks effectively. Therefore, the system is considered highly acceptable and suitable for implementation.

Table 4.2 To Evaluate

Criteria	Weighted Mean	Interpretation
Functional Suitability	**4.4**	Highly Agree
Reliability	**4.47**	Highly Agree
Performance Efficiency	**4.4**	Highly Agree
Usability	**4.6**	Highly Agree
Security	**4.6**	Highly Agree
Compatibility	**4.5**	Highly Agree
Maintainability	**4.4**	Highly Agree
Portability	**4.3**	Highly Agree
Overall Mean	**4.46**	Highly Agree

Based on the survey results, the system obtained a general weighted mean of 4.46, interpreted as “Highly Agree,” indicating a strong positive evaluation from the respondents. Functional suitability (4.4) confirmed that the system effectively performs schedule generation and conflict detection, while reliability (4.47) and performance efficiency (4.4) demonstrated stable and responsive system performance. Usability and security both received high ratings (4.6), showing that the system is user-friendly and ensures data protection. Compatibility (4.5), maintainability (4.4), and portability (4.3) further indicate that the system is adaptable and manageable across different environments.

These findings are consistent with the study of Reyes et al. (2023), which emphasized that web-based systems improve efficiency and provide real-time access to academic information. Similarly, Sarmiento et al. (2025) highlighted that systems evaluated using ISO 25010 standards achieve high levels of functionality, reliability, and user satisfaction. The results also align with Guirre and De Alba (2023), who found that automated scheduling systems significantly reduce conflicts and improve resource management, as well as Catubao et al. (2020) and Marquez et al. (2022), who demonstrated that computerized and web-based scheduling systems enhance efficiency and minimize errors in academic institutions.

Overall, the evaluation results validate that the system is efficient, reliable, and user-friendly, and capable of significantly improving the scheduling process of ACLC College of Taytay.

General Evaluation Questionnaire Results (Respondents)

This section presents the results of the general evaluation questionnaire answered by administrators, teachers, and students. The purpose of this evaluation is to assess the overall performance and effectiveness of the developed system based on the responses gathered from the respondents.

Table 4.3 Respondents

Question	Weighted Mean	Interpretation
1.The system makes scheduling tasks easier to perform.	**4.20**	Highly Agree
2.The system helps organize schedule information properly.	**4.24**	Highly Agree
3.The system improves the efficiency of the scheduling process in the school.	**4.40**	Highly Agree
4.The system helps reduce scheduling conflicts such as overlapping classes.	**4.24**	Highly Agree
5.The system provides consistent results across different uses.	**4.11**	Agree
6.The system allows users to access schedules anytime when needed.	**4.22**	Highly Agree
7.The system is accessible through a web browser without difficulty.	**4.11**	Agree

8.The system loads schedule information quickly.	**4.22**	Highly Agree
9.The system can be accessed using devices such as phones, laptops, or tablets.	**4.38**	Highly Agree
10.The system loads properly even with limited internet connection.	**4.18**	Agree
11.The layout and design of the system are visually clear and understandable.	**4.29**	Highly Agree
12.The menus and navigation options are easy to locate and use.	**4.29**	Highly Agree
13.The timetable or schedule format is easy to read and interpret.	**4.36**	Highly Agree
14.The system interface allows users to complete task without confusion.	**4.18**	Agree
15.The design of the system helps users quickly understand how to use its features.	**4.40**	Highly Agree
16.The system makes scheduling tasks faster compared to manual methods.	**4.20**	Highly Agree
17.The system minimizes errors in schedule creation and modification.	**4.38**	Highly Agree
18.The notification feature helps users stay informed about schedule updates.	**4.20**	Highly Agree

19.The system improves communication of scheduling information.	**4.18**	Agree
20.The system meets my needs in managing and controlling school schedules.	**3.98**	Agree
Overall Mean	**4.23**	**Highly Agree**

The results show that the system received a general weighted mean of 4.23, interpreted as "Highly Agree." This indicates that the respondents are highly satisfied with the system, particularly in improving scheduling efficiency, reducing conflicts, and providing a user-friendly interface. Overall, the system is considered effective and meets the needs of its users.

ISO Evaluation Results (ISO)

This section presents the results of the system evaluation based on the ISO 25010 Software Quality Model. The evaluation focuses on assessing the system's quality in terms of functionality, reliability, usability, performance efficiency, security, compatibility, maintainability, and portability based on the responses gathered from the respondents.

Table 4.4 Functional Suitability

Question	Weighted Mean	Interpretation
1.The system provides complete functions required for scheduling teachers, rooms, and sections.	**4.4**	Highly Agree
2.The system produces correct scheduling outputs based on given inputs.	**4.4**	Highly Agree
3.The system appropriately supports required scheduling tasks without missing features.	**4.4**	Highly Agree
General Weighted Mean	**4.4**	**Highly Agree**

In terms of Functional Suitability, all items received a mean of 4.4, showing that our system provides complete features, produces accurate outputs, and supports all required scheduling tasks effectively.

Table 4.5 Reliability

Question	Weighted Mean	Interpretation
4.The system maintains stable performance during continuous operation.	**4.4**	Highly Agree
5.The system efficiently handles multiple scheduling operations at the same time.	**4.6**	Highly Agree
6.The system preserves data consistency during interruptions or failures.	**4.4**	Highly Agree
General Weighted Mean	**4.47**	**Highly Agree**

For Reliability, the system demonstrated stable and consistent performance, with a high rating of 4.6 for handling multiple scheduling operations. This indicates that our system can operate continuously without failure and maintain data consistency.

Table 4.6 Performance Efficiency

Question	Weighted Mean	Interpretation
7.The system responds within acceptable time during scheduling operations.	**4.4**	Highly Agree
8.The system efficiently handles multiple user requests simultaneously.	**4.4**	Highly Agree
9.The system minimizes delays through optimized data processing.	**4.4**	Highly Agree
General Weighted Mean	**4.4**	**Highly Agree**

Under Performance Efficiency, all indicators received a mean of 4.4, showing that our system responds quickly and efficiently handles multiple user requests with minimal delay.

Table 4.7 Usability

Question	Weighted Mean	Interpretation
10.The system interface is intuitive for users performing scheduling tasks.	**4.4**	Highly Agree
11.The system provides clear and understandable feedback for user actions.	**4.8**	Highly Agree
General Weighted Mean	**4.6**	**Highly Agree**

In terms of Usability, our system was rated highly, especially in providing clear feedback to users (4.8), which indicates that users can easily understand and interact with the system.

Table 4.8 Security

Question	Weighted Mean	Interpretation
12.The system ensures secure authentication before granting access.	**4.6**	Highly Agree
13.The system enforces role-based access control for different user types.	**4.6**	Highly Agree
14.The system protects sensitive data from unauthorized access or modification.	**4.6**	Highly Agree
General Weighted Mean	**4.6**	**Highly Agree**

For Security, all indicators received 4.6, confirming that our system effectively implements authentication and role-based access control while protecting sensitive data from unauthorized access.

Table 4.9 Compatibility

Question	Weighted Mean	Interpretation
15.The system functions correctly across different web browsers.	**4.4**	Highly Agree
16.The system maintains consistent behavior across various device types.	**4.6**	Highly Agree
General Weighted Mean	**4.5**	**Highly Agree**

In Compatibility, our system performs consistently across different browsers and devices, with a high rating of 4.6, indicating flexibility and accessibility.

Table 4.10 Maintainability

Question	Weighted Mean	Interpretation
17.The system is designed with modular components for easier updates.	**4.2**	Highly Agree
18.The system codebase is structured to support efficient debugging and modification.	**4.6**	Highly Agree
General Weighted Mean	**4.6**	**Highly Agree**

For Maintainability, the results show that our system is well-structured and easy to modify, with ratings up to 4.6, meaning it can be updated and maintained efficiently.

Table 4.11 Portability

Question	Weighted Mean	Interpretation
19.The system adapts properly to different screen sizes and resolutions.	**4.2**	Highly Agree
20.The system runs consistently on devices such as phones, laptops, and tablets.	**4.4**	Highly Agree
General Weighted Mean	**4.3**	**Highly Agree**

Lastly, in Portability, our system adapts well to different screen sizes and devices, ensuring usability across multiple platforms.

CHAPTER V

Conclusions and Recommendations

Summary of Findings

The study revealed that the developed Web-Based Scheduling and Assignment System significantly improved the scheduling process at ACLC College of Taytay. Compared to the traditional manual method, the system made scheduling faster, more accurate, and more organized, reducing the chances of human error and overlapping schedules.

The system successfully implemented key features such as generated scheduling, conflict detection, user management, and role-based access. These features allowed administrators to efficiently assign teachers, rooms, and sections while preventing conflicts in time, room, and teacher availability.

In addition, the system provided convenience for teachers and students by allowing them to easily access their schedules through the "My Timetable" feature. The notification feature also ensured that users were updated in real time regarding schedule changes and announcements.

Based on the User Acceptance Testing (UAT), all test cases passed successfully, indicating that the system functions correctly and meets user requirements. Furthermore, the evaluation results using the ISO 25010 model showed high ratings in all criteria, with an overall mean interpreted as "Highly

Agree." This confirms that the system is functional, reliable, efficient, user-friendly, and secure.

Overall, the findings demonstrate that the system effectively addressed the problems of manual scheduling and improved the overall management of academic schedules.

Conclusions

Based on the findings of the study, the researchers conclude that the Web-Based Scheduling and Assignment System is an effective and reliable solution to the challenges of manual scheduling at ACLC College of Taytay.

The system successfully simplifies the scheduling process by using generated scheduling, reducing conflicts, and minimizing errors. It enhances efficiency by allowing faster schedule creation and easier data management for administrators. At the same time, it improves accessibility by enabling teachers and students to conveniently view their schedules anytime and anywhere.

Moreover, the system meets the ISO 25010 quality standards, particularly in terms of functionality, reliability, usability, security, and performance efficiency. This indicates that the system is not only technically sound but also highly acceptable to its users.

Therefore, the researchers conclude that the system is a practical, efficient, and user-friendly tool that can significantly improve scheduling management within the institution. It can also serve as a foundation for future enhancements and similar systems in other educational institutions.

Recommendations

- **Mobile Application Development**

  Develop a mobile application version of the system to improve accessibility, allowing users to conveniently access schedules anytime and anywhere.

- **Export Feature for Students and Teachers**

  Add an export functionality that enables students and teachers to download or save their schedules as files (e.g., PDF) or images for easy sharing and offline access.

- **Advanced Conflict Detection**

  Enhance the conflict detection feature to provide more detailed information, such as identifying which teacher, room, or section is causing the conflict.

- **Total Teaching Hours Calculation**

  Include a feature that automatically calculates and displays the total number of teaching hours assigned to each teacher.

- **Break Time Integration**

  Add a break time feature within the scheduling system to ensure proper time gaps between classes and avoid continuous or overloaded schedules.